Class 6
MATHEMATICS

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1. Number system

Worksheet – 1

Level – I

1) **D** 2) **B** 3) **C** 4) **D** 5) **B** 6) **A** 7) **B**
8) **C** 9) **B** 10) **C**

Level – II

1. $2367 + 3592 + 4572 = 10531$ is odd.
   Answer) **B**
2. Take 1,3,5 are odd numbers ⇒ Sum = 1 + 3 + 5 = 9 is odd
   Answer) **A**
3. Answer) **D**
4. $(1062 - 1052) - 1 = 10 - 1 = 9$
   Answer) **C**
5. Least natural number is 1⇒ Least whole number is 0
   Sum of 0 and 1 is 1
   Answer) **A**
6. Answer) **A**
7. First five whole numbers are 0,1,2,3,4⇒Product = 0×1×2×3×4 = 0
   Answer) **D**
8. Successor of 105 = 106 Predecessor of 106 = 105
   Difference = 106 -105 =1
   Answer) **D**
9. Even
   Answer) **A**
10. largest whole number preceding $10^8 + 1 = (10^8+1)-1 = 10^8 + 1 - 1 = 10^8 = 10^8 \times 1$
    Answer) **C**
11. even + even + even = even
    Answer) **A**
12. Let 'a' be any number
    Successor of a is a +1
    Predecessor of a is a -1
    Difference = a +1- (a -1) = 2
    Answer) **2**

Level – III

13) **B,C** 14) **B,D**
15) **C** 16) **C** 17) **C** 18) **B** 19) **a-r, b-p, c-q, d-t**
20) **1235** 21) **0**

Worksheet – 2

Level – I

1) **B** 2) **C** 3) **A** 4) **B** 5) **C** 6) **C**
7) **C** 8) **C** 9) **C** 10) **D** 11) **B**

Level – II

1) **D** 2) **C** 3) **C** 4) **A** 5) **A** 6) **C**

7) **C**

8) (3,5), (5,7) , (11,13) , (17,19) , (29,31) , (41,43) , (59,61), (71,73)

Total number of twin primes = 8

Answer) **B**

9) **B**

10) 81, 82, 84, 85, 86, 87, 88

Answer) **A**

11. Prime factors of 2019 is 3 and 673.

Answer) **2**

12. 1+ 2 + 3 + 6 =12 = 2(6)

Answer) **6**

Level – III

13) B,D 14) A,C 15) C 16) A 17) B 18) C

19) a-s, b-p, c-s, d-q

20) 43

Worksheet – 3

Level – I

1.

$2|\underline{28}$

$2|\underline{14}$

7

$\therefore 28 = 2^2 \times 7$

Answer) **C**

2.

$2|\underline{24}$

$2|\underline{12}$

$2|\underline{6}$

3

$\therefore 24 = 2^3 \times 3$

Answer) **D**

3. $14 = 2^1 \times 7^1$No of divisors $= (1+1)(1+1) \Rightarrow \times 2 = 4$

Answer) **C**

4. $70 = 2^1 \times 5^1 \times 7^1$No of divisors$= (1+1)(1+1)(1+1) = 2 \times 2 \times 2 = 8$

Answer) **B**

5. $28 = 2^2 \times 7^1$Sum of divisors$= \left(\frac{2^{2+1}-1}{2-1}\right)\left(\frac{7^{2+1}-1}{7-1}\right) = \frac{7}{1} \times \frac{48}{6} = 56$

Answer) **C**

6. $42 = 2^1 \times 3^1 \times 7^1$Sum of divisors$= \left(\frac{2^{1+1}-1}{2-1}\right)\left(\frac{3^{1+1}-1}{3-1}\right)\left(\frac{7^{1+1}-1}{7-1}\right) = \frac{3}{1} \times \frac{8}{2} \times \frac{48}{6} = 96$

Answer) **B**

7. $20 = 2^2 \times 5^1$ product of divisors of $20 = 20^{\frac{1}{2}(2+1)(1+1)} 20^{\frac{1}{2}\times 3\times\times 2} 20^3$
Answer) **B**

8. $24 = 2^3 \times 3^1$ product of divisors $24 = 24^{\frac{1}{2}(3+1)(1+1)} = 24^{\frac{1}{2}\times 4\times 2} = 24^4$
Answer) **D**

9) **C**

Level – II

1. $96 = 2^5 \times 3^1$ No of divisors $= (5+1)(1+1) = 6 \times 2 = 12$
Answer) **A**

2. $280 = 2^3 \times 5^1 \times 7$ No of divisors $= (3+1)(1+1)(1+1) = 4 \times 2 \times 2 = 16$
Answer) **B**

3. $16 = 2^4$ Sum of divisors $= \frac{2^{4+1}-1}{2-1} = 31$
Answer) **B**

4. $26 = 2^1 \times 13^1$ Sum of divisors $= \left(\frac{2^{1+1}-1}{2-1}\right)\left(\frac{13^{1+1}-1}{13-1}\right) = \frac{3}{1} \times \frac{168}{12} = 42$
Answer) **B**

5. $5005 - 5 \times 7 \times 11 \times 13$
Answer) **C**

6. $18 = 2^1 \times 3^2$ sum of divisors of $18 = 13 \times k \left(\frac{2^{1+1}-1}{2-1}\right)\left(\frac{3^{1+1}-1}{3-1}\right) = \frac{3}{1} \times \frac{26}{2} = 13 \times k$
$13 \times 3 = 13 \times k \therefore k = 3$
Answer) **C**

7. $75 = 3^1 \times 5^2$ product of divisors of $75 = 75^{\frac{1}{2}(1+1)(2+1)} = 75^{\frac{1}{2}\times 2\times 3} = 75^3$
Answer) **B**

8. $128 = 2^1$ product of divisors of $128 = 128^{\frac{1}{2}(1+1)} = 128^4$
Answer) **C**

9. $286 = 2 \times 11 \times 13$
Answer) **B**

10. $1080 = 2^3 \times 3^3 \times 5^1$ no of divisors $= (3+1)(3+1)(1+1) = 4 \times 4 \times 2 = 32$
Answer) **32**

11. $512 = 2^9$ product of divisors $= 512^{\frac{1}{2}(9+1)} = 512^5 = (2^9)^5 = 2^{45}$
Answer) **1**

12. $1024 = 2^4 \times 5^1 \times 13^1$ sum of divisors$= \left(\frac{2^{4+1}-1}{2-1}\right)\left(\frac{5^{1+1}-1}{5-1}\right)\left(\frac{13^{1+1}-1}{13-1}\right) = \frac{31}{1} \times \frac{24}{4}$
$\frac{168}{12} = 2604$
Answer) **1**

Level – III

13. $18 = 2^1 \times 3^2$ No of divisors $= (1+1)(2+1) = 2 \times 3 = 6$
Answer) **C,D**

14. $24 = 2^3 \times 3^1$ No of divisors $= \left(\frac{2^{3+1}-1}{2-1}\right)\left(\frac{3^{1+1}-1}{3-1}\right) = \frac{15}{1} \times \frac{8}{2} = 60$
    Answer) **B,D**
15. Answer) **A**
16. $91 = 13^1 \times 7$
    Answer) **C**
17. $97 = 97^1$product of divisors $= 97^{\frac{1}{2}(1+1)} = 97^{\frac{1}{2}(2)}97$
    Answer) **C**
18. $343 = 7^3$ product of divisors $= 343^{\frac{1}{2}(3+1)} = 343^{\frac{1}{2}\times 4} = 343^2 = (7^3)^2 = 7^6$
    Answer) **D**
19. a-q, b-r, c-p, d-s
20. $150 = 2^1 \times 3^1 \times 5^2$ product of divisors $150^{\frac{1}{2}(1+1)(1+1)(2+1)} = 150^{\frac{1}{2}(2)(2)(3)} = 150^6$
    Answer)150^6

Worksheet – 4

Level – I

1) **C** 2) **A** 3) **C** 4) **C**

5. Sum of all digits is divisible by ' 9 '.
   Answer) **D**
6. The number which is divisible by 2&3.
   Answer) **B**

7) **A** 8) **D** 9) **B** 10) **C**

Level – II

1. A number is divisible by 2 if and only if the digit at its unit's place either 0 or 2 (or) 4 (or) 6 (or) 8 .
   Answer) **C**
2. Using divisibility rule 2
   Answer) **C**
3. Using divisibility rule 3 .
   Answer) **B**
4. 5 + 4 + 5 + 4 = 18 is divisible of 3
   Answer) **C**
5. 7 + 2 + 3 = 12 is divisible by 3
   Answer) **C**
6. Using divisibility rule 10
   Answer) **A**
7. 576 - by option verification.

Answer) **A**

8. Using divisibility rule 2 and 3 .

Answer) **C**

9. $2 + 3 + 5 + 8 = 18$ is divisible by 9

Answer) **D**

10. $1 + 6 + 6 + 8 = 21$ is divisible by 3 but not 9 .

Answer) **C**

11) **1**

12) **2**

Level – III

13. $1 + 2 + 3 + 4 + 5 = 15$ which is divisible by 3

$4 + 5 + 3 = 12$ which is divisible by 3

$3 + 6 + 9 + 0 = 18$ is divisible by 3

Answer) **A,B,C**

14. Using divisibility rules of 3 and 9&2

Answer) **A,B,D**

15. Answer) **C**

16. 320418; $3 + 2 + 0 + 4 + 1 + 8 = 18$ which is divisible by 9 .

Answer) **D**

17. 120360 which is divisible by both 2 and 3 .

Answer) **A**

18. 112233

$1 + 1 + 2 + 2 + 3 + 3 = 12$ which is divisible by 3 but not 9

Answer) **C**

19. Answer) **a-p,q,r,s; b-p,q,r; c-t; d-p,q,r,s,t**

20. Using divisibility rule by 5 .

Answer) **5**

Worksheet – 5

Level – I

1. Answer) **C**

2. using Divisibility rule of 8

Answer) **C**

3. Answer) **C**

4. using Divisibility rule of 11

Answer) **B**

5. using Divisibility rule of 4

   Answer) **C**

6. Answer) **B**

7. using Divisibility rule of 11

   Answer) **C**

8. Answer) **D**

9. using Divisibility rule of 4 and 8 .

   Answer) **C**

Level – II

1. Sum of odd places - Sum of even places

   $1 + x = 24$

   put $x = 1$ then $24 - 2 = 22$ which is divisible by 11

   Answer) **B**

2. 75^{*} is divisible by 4

   the smallest number replace * by ' 2 ' then 752 is divisible by 4

   Answer) **C**

3. use Divisibility rule of 11

   Sum of odd places - Sum of even places = 0 (or) divisible by 11

   Answer) **C**

4. Last two digits divisible by 4

   Answer) **A**

5. Sum of odd places - Sum of even places = 0 then 311113 is divisible by 11

   Answer) **A**

6. using Divisibility rule of 8

   Answer) **A**

7. Last 3 digits is divisible by 8

   Answer) **A**

8. using Divisibility rule of 4

   Answer) **A**

9. using Divisibility rule of 4

   Answer) **C**

10. Sum of odd places - Sum of even places = 11 which is divisible by 11

    Answer) **D**

11. Answer(**11**

Level – III

12. using Divisibility rule of 8

    Answer) **B, C**

13. using Divisibility rule of 4

    Answer) **A, D**

14. Answer) **A**

15. Answer) **A**

16. Answer) **B**

17. 79528721 + x is divisible by 4

    put x = 7 then 79528728 is divisible by 4

    Answer) **C**

18. Answer) **a-q,r,s; b-s; c-p; d-q,r,s**

19. Answer) **11**

Worksheet – 6

Level – I

1. Answer) **B**

2. Answer) **C**

3. Answer) **C**

4. $10 = 2 \times 5, 30 = 2 \times 5 \times 3 \Rightarrow$ H.C.F $= 2 \times 5 = 10$

   Answer) **C**

5. $4 = 2^2, 12 = 2^2 \times 3 \Rightarrow$ G.C.D $= 2^2 = 4$

   Answer) **B**

6. $4 = 2^2, 8 = 2^3 \Rightarrow 16 = 2^4 \Rightarrow$ G.C.F $= 2^2 = 4$

   Answer) **C**

7. $24 = 2^3 \times 3$

   Answer) **B**

8. Answer) **C**

9. 2 | 4,12

   2 | 2,6

   1,3

   L.C.M $= 2 \times 2 \times 1 \times 3 = 12$

   Answer) **A**

10. 2 | 8,12,18

    2 | 4,6,9

3 | 2,3,9

2,1,3

L.C. $M = 2 \times 2 \times 3 \times 2 \times 1 \times 3 = 72$

Answer) **B**

Level – II

1. $x = 2 \times 2 \times 3 \times 3 \times 5 = 2^2 \times 3^2 \times 5 \Rightarrow y = 2 \times 3 \times 3 \times 3 = 2^1 \times 3^3$

   $L.C.M = 2^2 \times 3^3 \times 5 = 4 \times 27 \times 5 = 540$

   Answer) **C**

2. Answer) **C**

3. $x = 5 \times 7 \times 9 = 5 \times 7 \times 3^2, y = 5 \times 3 \times 11 = 5 \times 3 \times 11 \Rightarrow H.C.F = 5 \times 3 = 15$

   Answer) **B**

4. $p = a \times m \times \gamma, q = b \times m \times \gamma \times 2 \Rightarrow H.C.F = m \times \gamma = m\gamma$

   Answer) **B**

5. $101 = 7 \times 13,573 = 3 \times 191 \Rightarrow 1079 = 13 \times 83 \Rightarrow H.C.F = 1$

   Answer) **A**

6. $45 = 3^2 \times 5^1, 120 = 3^1 \times 5^1 \times 2^3 \Rightarrow$ G.C. $D = 3^1 \times 5^1 = 15$

   Answer) **A**

7. $9 = 3^2, 36 = 3^2 \times 2^2 \Rightarrow$ $H.C.F = 3^2 = 9$

   Answer) **B**

8. $a = p \times p \times q \times r \times s = p^2 \times q \times r^1 \times s^1 \Rightarrow b = p \times q \times q \times s = p^1 \times q^2 \times s^1$

   $G.C.D = p \times q \times s$

   Answer) **B**

9. Answer) **C**

10. $72 = 2^3 \times 3^2, 90 = 2^1 \times 3^2 \times 5^1 \Rightarrow 99 = 3^2 \times 11 \Rightarrow G.C.D = 3^2 = 9$

    Answer) **B**

11. $576 = 2^6 \times 3^2, 720 = 2^4 \times 3^2 \times 5^1 \Rightarrow L.C.M = 2^6 \times 3^2 \times 5^1$

    Answer) **C**

12. $p = 3^2 \times 17 \times 19, q = 3^2 \times 2 \times 19 \Rightarrow H.C.F = 3^2 \times 19$

    Answer) **3^2x19**

13. $18 = 2^1 \times 3^2, 27 = 3^3 \Rightarrow L.C.M = 2^1 \times 3^3 = 54$

    Answer) **54**

14. $12 = 2^2 \times 3^1, 18 = 2 \times 3^2 \Rightarrow G.C.D = 2 \times 3 = 6$

    Answer) **A,B,C,D**

15. Answer) **C**

16. Answer) **D**

17. $144 = 2^4 \times 3^2, 180 = 2^2 \times 3^2 \times 5^1, 198 = 2 \times 3^2 \times 11$

H.C.F $= 2 \times 3^2 = 2 \times 9 = 18$

Answer) **B**

18. $P = 2 \times 3 \times 5 \times 7, Q = 2^2 \times 3 \times 5 \times 7 \Rightarrow$ H.C.F $= 2 \times 3 \times 5 \times 7 = 210$

Answer) **A**

19. $x = 2^2 \times 3 \times 5 \times 7$, $y = 3 \times 5 \times 19 \Rightarrow$ H.C.F $= 3 \times 5 = 15$

Answer) **B**

20. Answer) **a-q;b-s;c-p;d-r**

21. Answer) 6

Worksheet – 7

Level – I

1. $a \times b = \dot{H}.C.F \times L.C.M \Rightarrow H.C.F = \frac{12 \times 15}{60} = 3$

Answer) **D**

2. L.C.M $= \frac{a \times b}{H.C.F} = \frac{64}{16} = 4$

Answer) **B**

3. $a \times b = L.C.M \times H.C.F \Rightarrow b = \frac{L.C.M \times H.C.F}{A} = \frac{6 \times 36}{12} = \frac{216}{12} = 18$

Answer) **B**

4. H.C.F of fractions $= \frac{\text{H.C.F of Numerators}}{\text{L.C.M of Denominators}} \Rightarrow$ H.C.F of $\frac{1}{2}, \frac{1}{3}, \frac{1}{4} = \frac{1}{12}$

Answer) **C**

5. Answer) **A**

6. L.C.M of fractions $= \frac{\text{L.C.M of Numerators}}{\text{H.C.F of Denominators}} \Rightarrow$ L.C.M of $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ is $\frac{1}{1} = 1$

Answer) **C**

7. Answer) **D**

8. Product of two numbers = L.C.M × H.C.F

Answer) **D**

9. L.C.M of $\frac{2}{4}, \frac{1}{4}, \frac{1}{8}$ is $\Rightarrow \frac{\text{L.C.M of (2,1,1)}}{\text{H.C.F of (4,4,8)}} = \frac{2}{4}$

Answer) **C**

10. H.C.F of $2, \frac{1}{3}, \frac{1}{4}$ is $\frac{\text{H.C.F of (2,1,1)}}{\text{L.C.M of (1,3,4)}} = \frac{1}{12}$

Answer) **B**

Level – II

1. HCF of $\frac{8}{15}, \frac{16}{125} = \frac{\text{HCF of (8,16)}}{\text{LCM of (15,125)}} = \frac{8}{375}$

Answer) **C**

2. HCF of (1,2,3,4) = 1; LCM of (2,3,4,5) = 60 ⇒ $HCF = \frac{1}{60}$

   Answer) **D**

3. Product of numbers = 7 × 105 = 735

   Answer)

   A

4. G.C.D of (3,5,7) = 1, L.C.M of (16,12,18) = 144 ⇒ G.C.D is $\frac{1}{144}$

   Answer) **A**

5. L.C.M of (8,16) = 16, H.C.F of (15,125) = 5 ⇒ L.C.M = $\frac{16}{5}$

   Answer) **B**

6. L.C.M of (1,2,3,4) is 12 , H.C.F of (2,3,4,5) is 1 ⇒ L.C.M = $\frac{12}{1} = 12$

   Answer) **D**

7. $\text{L.C.M} = \frac{320}{HCF} = \frac{320}{4} = 80$

   Answer) **C**

8. $\text{G.C.D} = \frac{726}{66} = 11$

   Answer) **B**

9. H.C.F of (21,27,63) = 3, L.C.M of (10,10,10) = 10 ⇒ H.C.F = $\frac{3}{10}$

   Answer) **B**

10. Product of two numbers = $LCM \times HCF$

    $a \times b = LCM \times HCF; b = \frac{78 \times 13}{26} = 39$

    Answer) **39**

11. Answer) **C, D**

12. HCF of (2,3,5) is 1 , LCM of (7,14,3) is 42 ⇒ $HCF = \frac{1}{42}$

    Answer) **B, D**

13. Answer) **A**

14. $b = \frac{HCF \times LCM}{a} = \frac{16 \times 160}{32} = 80$ 14. $HCF = \frac{a \times b}{LCM} = \frac{248 \times 868}{1736} = 124$

    15. $\text{LCM} = \frac{a \times b}{HCF} = \frac{1850}{37} = 50$ 16. $b = \frac{L.C.M \times H.C.F}{a} = \frac{31 \times 1488}{186} = 248$

    Answer) **A**

15. Answer) **D**

16. Answer) **A**

17. a) HCF of $\frac{1}{2}, \frac{1}{3}$ is $\frac{1}{6}$ b) HCF of $\frac{2}{7}, \frac{4}{21}$ is $\frac{2}{21}$

    c) L.C.M of $\frac{1}{2}, \frac{1}{3}$ is 1 Class 6 – Mathematics

    d) L.C.M of $\frac{2}{7}, \frac{5}{21}$ is $\frac{10}{7}$

    Answer) **a-r; b-s; c-p;d-q**

18. HCF of two co-primes is 1 $\Rightarrow k = HCF$ of $(p, q) \Rightarrow k = 1$

    Answer) **K = 1**

19. L.C.M of (4,8,2) is 8 ; H.C.F of (3,5,7) is 1 ; L.C.M = 8

    Answer) **8**

2. INTEGERS

Worksheet – 1

Level – I

1) **B** 2) **C** 3) **D** 4) **C** 5) **C** 6) **D** 7) **A**

8) **B** 9) **C** 10) **A**

Level – II

1) **B** 2) **A** 3) **A** 4) **B** 5) **A**

6. $12 + |-12| \square 2 + |+3| \Rightarrow 12 + 12 \square 2 + 3 \Rightarrow 24 \square 5 \ \therefore 24 > 5$

   Answer) **B**

7. $|-1| + |+1| + |-3| = 1 + 1 + 3 = 5$

   Answer) **A**

8. $8 + |+4| = 8 + 4 = 12$

   Answer) **C**

9. $|-5| - |6| + |-8| = 5 - 6 + 8 = -1 + 8 = 7$

   Answer) **C**

10. Answer) **C**

11. Let $a \in N \Rightarrow 1 \times a = a, 1 + a = 1 + a \Rightarrow a < 1 + a \Rightarrow \therefore$ one number must be ' 1 '

    Answer) **A**

12. Let $n^3 = 27 = 3^3 \ \therefore n = 3$ is odd

    $n^2 = 3^2 = 3 \times 3 = 9$ is odd

    Answer) **C**

13. Answer) **A,B,C**

14. Answer) **A,B,C**

15. Answer) **A**

16. Answer) **B**

17. $100| + |-30| - |676| = 100 + 30 + 676 = 130 + 676 = 806$

    Answer) **B**

18. Answer) **A**

19. Answer) **a-r;b-p; c-q; d-s**

20. Answer)**1**

21. Answer) **B,C,D**

22. Answer) **B,C**

23. Answer) **A**

24. Answer) **a-r; b-p,q,r,s,t; c-t; d-q,r,s,t**

Worksheet – 2

Level – I

1) **A** 2) **D** 3) **A** 4) **A** 5) **B** 6) **A** 7) **D**

8) **C** 9) **D** 10) **C** 11) **C** 12) **A** 13) **C**

Level – II

1. $-208 + 40 - 4 - 10 + 15 = -167$

   Answer) **A**

2. $-2 \times 5 \times -30 = +300$

   Answer) **B**

3. $-7 - 8 + 25 = 10$

   Answer) **D**

4. Answer) **C**

5. $\frac{35}{-7} = -5$

   Answer) **B**

6. $15 \times -25 \times -4 \times -10 = -15{,}000$

   Answer) -15000

7. $-107 - (-20) - 5 = -92$

   Answer) **B**

8. $-17 \times -29 = 493$

   Answer) **A**

9. $7 \times (48) = 336$

   Answer) **B**

10. $103 - 57 \times -19 + 57 = 114$

    Answer) **C**

11. $8 \times 53 \times 125 = 53{,}000$

    Answer) **53000**

12. $22 \times (3) \times (-2) \times (-1) = 132$

    Answer) **132**

13. $60 + x = -300 \Rightarrow x = -360$

    Answer)**–360**

14. Answer) **A,C**

15. $a - (-8) = 15 \Rightarrow a = 15 - 8 = 7$

    Answer) **B,D**

16. $3 - k = -5 \Rightarrow k = 3 + 5 = 8$

    Answer) **A,D**

17. Answer) **A**

18. Answer) **A**

19. $-6 \times -15 = 6 \times 15 = 90$

    Answer) **D**

20. $-75 \times 3 = 3 \times -75 = -225$

    Answer) **D**

21. $\frac{-35}{-7} \times -42 = 5 \times -42 = -40$

    Answer) **D**

22. Answer) a-s; b-r; c-q;d-p

23. $1 - 1 + 1 - 1 + \cdots$..101 times $= 1$

    Answer) **1**

24. $-9 \times -7 = 63$

    Answer) **A,B,D**

25. $\frac{-68}{-34} \times 2 = 2 \times 2 = 4$

    Answer) **A,D**

26. Answer) **A**

27. $-8 \times (-7 + 5) = -8 \times -2 = 16$

    Answer) **16**

Worksheet – 3
Level – I

1. Answer) **A**
2. Answer) **B**
3. Answer) **D**
4. Answer) **A**
5. Answer) **C**

6. $\frac{5}{10} = \frac{\mathrm{m}}{30} \Rightarrow \mathrm{m} = \frac{5}{10} \times 30 = 15$

   Answer) **C**

7. $\frac{4}{12} = \frac{2}{6}$

   Answer) **A**

8. $\frac{15}{25} = \frac{3}{5}$

   Answer) **D**

9. $\frac{4\times7\times4}{7} = \frac{32}{7}$

   Answer) **D**

10. Answer) **D**

11. Answer) **D**

12. Answer) **A**

Level – II

1. By cross multiplication $\Rightarrow 2 \times 4 < 3 \times 3 \Rightarrow \frac{2}{3} < \frac{3}{4}$

   Answer) **B**

2. By option verification $\frac{1}{3}$

   Answer) **A**

3. $\frac{1}{3}$ and $\frac{3}{5} \Rightarrow 1 \times 5 < 3 \times 3 \Rightarrow 5 < 9 \ \therefore \frac{1}{3} < \frac{3}{5}$

   Answer) **C**

4. $\frac{200}{1000} = \frac{1}{5}$

   Answer) **A**

5. LCM of 3,2,6 is 6 $\Rightarrow$ Now $\frac{2}{3} = \frac{2\times2}{3\times2} = \frac{4}{6}; \frac{1\times3}{2\times3} = \frac{3}{6}; \frac{1}{6}$

   Compare the numerators $\Rightarrow \frac{1}{6} < \frac{3}{6} < \frac{4}{6}$ So, $\frac{1}{6}, \frac{1}{2}, \frac{2}{3}$

   Answer) **A**

6. $\frac{2\times9+1}{19} = \frac{18+1}{19} = \frac{19}{19} = 1$

   Answer) **1**

7. $\frac{10}{100} = \frac{1}{10}$

   Answer) 1/10

8. L.C.M of 5,3,7 = 105

   Now $\frac{2\times21}{5\times21} = \frac{42}{105}; \frac{1\times35}{3\times35} = \frac{35}{105}; \frac{3\times15}{7\times15} = \frac{45}{105}$

   Compare the numerators

   $\frac{42}{105} > \frac{45}{105} > \frac{35}{105}$ So, $\frac{3}{7} > \frac{2}{5} > \frac{1}{3}$

   Answer) **C**

9. $\frac{-3\times2}{2\times2}, \frac{3\times2}{2\times2}, \frac{11}{4}, \frac{5\times2}{2\times2} \Rightarrow \frac{-6}{4}, \frac{6}{4}, \frac{11}{4}, \frac{10}{4} \Rightarrow \frac{11}{4}$ is greatest.

Answer) **B**

10. Compare numerators

Answer) **A**

11. By option verification

Answer) **B**

12. By option verification

Answer) **B**

13. By option verification

Answer) **C**

13. $3\frac{2}{3} = \frac{11}{3} \Rightarrow$ numerator $= 11 \Rightarrow$ But given $8 + K = 11 \Rightarrow K = 3$

Answer) **3**

14. Option verification 15. Option verification

Answer) **B,C**

15. Answer) **A,B,C**

16. Answer) **D**

17. Answer) **D**

18. $\frac{5}{10} = \frac{m-2}{30} \Rightarrow m - 2 = 15 \Rightarrow m = 17$

Answer) **B**

19. $\frac{2\times m+3}{19} = \frac{37}{19} = 1\frac{18}{19}$

Answer) **D**

20. $\frac{m-10}{21} = \frac{17-10}{21} = \frac{7}{21} = \frac{1}{3}$

Answer) **A**

21. a) $\frac{35}{100} = \frac{7}{20}$

b) $\frac{75}{200} = \frac{3}{8}$

c) $\frac{16}{24} = \frac{2}{3}$

d) $\frac{250}{3000} = \frac{1}{12}$

Answer) **a-p, b-q, c-r, d-s**

22. $7\frac{8}{9} = \frac{71}{9}$; $70 + x = 71 \Rightarrow x = 1$

Answer) **A**

23. Option verification

Answer) **C,D**

24. $\frac{20}{60} = \frac{1}{3}$

Answer) **B**

25. $\frac{15}{60} = \frac{1}{4}$

Answer) **C**

26. $\frac{25}{60} = \frac{5}{12}$

Answer) **D**

27. i) $3 - \frac{2}{6} = \frac{18-2}{6} = \frac{16}{6} = \frac{8}{3}$

ii) $\frac{a}{b} - \frac{b}{a} = \frac{a^2-b^2}{ab}$

Answer) **A**

28. $\frac{5}{9} - \frac{1}{9} = x \Rightarrow x = \frac{4}{9}$

Answer) **A**

3. RATIONAL NUMBERS

Worksheet – 1

Level – I

1) **B** 2) **B** 3) **B** 4) **C** 5) **C** 6) **A**

7) **A** 8) **A** 9) **A** 10) **C** 11) **B**

Level – II

1. A Completed $\frac{3}{7}$ part of work ⇒ B completed $\frac{4}{9}$ part of work.

   $A \to \frac{3}{7} \times \frac{9}{9} = \frac{27}{63} \Rightarrow B \to \frac{4}{9} \times \frac{7}{7} = \frac{28}{63}$

   ∴ A complete lesser part

   Answer) **B**

2. Left out cake was $1 - \frac{5}{7} = \frac{2}{7}$

   Answer) **C**

3. $-\frac{5}{19} = \frac{5}{19} \times \frac{6}{6} = \frac{-30}{114}$

   Answer) **C**

4. $\frac{-12}{7} = \frac{-12}{7} \times \frac{9}{9} = \frac{-108}{63}$

   Answer) **D**

5. $\frac{-40/8}{56/8} = \frac{-5}{7}$

   Answer) **B**

6. $\frac{-3}{2} < \frac{-3}{4} \Rightarrow -3 \times 4 < -3 \times 2$ and $-k \times 4 < -3 \times 2 \Rightarrow k = 3$

   Answer) **3**

7. $-\frac{63}{56} = \frac{-K}{8} \Rightarrow K = 9$

   Answer) **9**

8. $\frac{-8}{147} = \frac{-84/21}{147/21} = \frac{-4}{7}$

   Answer) **A**

9. $\frac{3 \times x}{2 \times y} = \frac{48}{72}$; $\frac{x}{y} = \frac{4}{9}$

   Answer) **C**

10. 3rd multiple of 4 is 12 ⇒ 8 th multiple of 2 is 16 ⇒ Now $\frac{12}{16} = \frac{12}{16} \times \frac{6}{6} = \frac{72}{96}$

    Answer) **A**

11. Monthly salary is Rs. 19,200

    He spent 2500 + 3,800 + 4000 + 41100

Saving is $19{,}200 - 14{,}400 = 4800$

Fraction of savings $= \frac{4800}{19{,}200} = \frac{1}{4}$

Answer) **C**

12. $\frac{72}{24\times100} = \frac{3}{100}$

Answer) **B**

13. Answer) **3**

Level – III

14. Answer) **A,B,D**
15. Answer) **A,B,C**
16. Answer) **A**
17. Answer) **A**
18. $\frac{5\times(-6)\times(-9)}{(-3)\times9\times4} = \frac{-5}{2}$

Answer) **B**

19. $\frac{23}{29} = \frac{x}{87} \Rightarrow x = 69$

Answer) **D**

20. $\frac{(-2)\times(-3)\times(-4)}{(-6)(-9)\times(-1)} = \frac{4}{9}$

Answer) **B**

21) Answer) **a-p,q,r;b-p,q,r;c-p,q,r,t;d-q,s**

22. 16th multiple of 2 is $32 \Rightarrow 2^{nd}$ multiple of 4 is 8 $\therefore$ quotient is 4

Answer) **4**

23) Answer) **A,C**

24) Answer) **A,C**

25) Answer) **A**

26) Answer) **A**

27) Answer) **C**

28) Answer) **B**

29) Answer) **D**

Worksheet – 2

Level – I

1. $\frac{1}{7} + \frac{15}{7} = \frac{16}{7}$

Answer) **A**

2. $\frac{7}{12}+\frac{9}{12}=\frac{3}{12}=\frac{19}{12}$

   Answer) **B**

3. $1-\frac{5}{12}=\frac{12-5}{12}=\frac{7}{12}$

   Answer) **B**

4. $\frac{15}{15}=1$

   Answer) **A**

5. $\frac{1}{5}-\frac{7}{5}=\frac{-6}{5}$

   Answer) **A**

6. $\frac{1}{2}-\frac{3}{2}=-1$

   Answer) **C**

7. $\frac{5}{7}-\frac{3}{7}=\frac{2}{7}$

   Answer) **D**

8. $\frac{1}{3}+\frac{2}{6}+\frac{5}{6}=\frac{2+2+5}{6}=\frac{9}{6}=\frac{3}{2}$

   Answer) **A**

9. $\frac{7}{3}+\frac{5}{3}=\frac{12}{3}=4$

   Answer) **C**

Level – II

1. $\frac{13}{4}-\frac{11}{8}=\frac{26-11}{8}=\frac{15}{8}=1\frac{7}{8}$

   Answer) **A**

2. $\frac{-5}{7}+\frac{2}{5}=\frac{-25+14}{35}=\frac{-11}{35}$

   Answer) **D**

3. $\frac{1}{2}+\left(\frac{-2}{3}+\frac{3}{4}\right)=\frac{1}{2}+\left(\frac{-8+9}{12}\right)=\frac{1}{2}+\frac{1}{12}=\frac{6+1}{12}=\frac{7}{12}$

   Answer) **B**

4. $\frac{4}{7}-\frac{8}{9}-\frac{13}{7}+\frac{17}{21}=\frac{-9}{7}+\frac{17}{21}-\frac{8}{9}=\frac{-27+17}{21}-\frac{8}{9}=\frac{-10}{21}-\frac{8}{9}=\frac{-30-56}{63}=\frac{-86}{63}$

   Answer) **A**

5. $-8-\left(\frac{-17}{9}\right)=\frac{-72+17}{9}=\frac{-55}{9}$

   Answer) **C**

6. $\frac{5}{3}-\frac{8}{3}+\frac{11}{2}-\frac{9}{8}-\frac{7}{2}+\frac{17}{8}=\frac{-3}{3}+\frac{4}{2}-\frac{9}{8}+\frac{17}{8}$

   $=-1+2-\frac{9}{8}+\frac{17}{8}=1-\frac{9}{8}=\frac{-1}{8}+\frac{17}{8}=\frac{16}{8}=2$

Answer) **2**

7. $\frac{50}{39}+\frac{-5}{13}-\left(\frac{-4}{39}\right)=\frac{50}{39}+\frac{-15+4}{39}=\frac{-11}{39}+\frac{50}{39}=1$

Answer) **1**

8. $\frac{9}{16}+\frac{7}{12}=\frac{27+28}{48}=\frac{55}{48}$

Answer) **D**

9. $B-A+D-C=\frac{3}{5}+\frac{2}{15}+\frac{3}{4}+\frac{15}{16}=\frac{9+2}{15}+\frac{12+15}{16}=\frac{11}{15}+\frac{27}{16}=\frac{176+405}{240}=\frac{581}{240}$

Answer) **A**

10. $x=\frac{12}{27}-\frac{5}{27}=\frac{7}{27}$

Answer) **A**

11. $\frac{-8}{15}-\left(\frac{3}{4}+\frac{2}{5}\right)=-\frac{8}{15}-\left(\frac{15+8}{20}\right)=-\frac{8}{15}-\frac{23}{20}=\frac{-32-69}{60}=\frac{-101}{60}$

Answer) **A**

12. $R=-(P+Q)=-\left(\frac{7}{8}-\frac{9}{2}+\frac{2}{3}+\frac{3}{4}\right)$

$=-\left(\frac{7-36}{8}+\frac{8+9}{12}\right)=-\left(\frac{-29}{8}+\frac{17}{12}\right)=-\left(\frac{-87+34}{24}\right)=-\left(\frac{-53}{24}\right)=\frac{53}{24}$

Answer) **C**

13. $\frac{5}{9}-x=\frac{2}{3}\longrightarrow x=\frac{5}{9}-\frac{2}{3}=\frac{5-6}{9}=\frac{-1}{9}$

Answer) **A**

14. $\frac{a}{b}+\frac{a}{c}+\frac{a}{d}+\frac{a}{e}=\frac{3567}{10}+\frac{3567}{100}+\frac{3567}{1000}+\frac{3568}{10000}=356.7+35.67+3.567+0.3567=396.2937$

Answer) **B,D**

15. $\frac{33}{4}-\frac{17}{6}=\frac{99-34}{12}=\frac{65}{12}=5\frac{5}{12}$

Answer) **A,C**

16. (i) $2+\frac{5}{9}=\frac{23}{9}$; (ii) $\frac{4}{5}+\frac{2}{5}=\frac{6}{5}\neq 1$

Answer) **C**

17. $\frac{2}{3}+\frac{5}{3}=\frac{7}{3}$

Answer) **C**

18. $\frac{-6}{7}+0=\frac{-6}{7}$

Answer) **C**

19. $\frac{-19}{3}-\frac{2}{3}=\frac{-21}{3}=-7$

Answer) **A**

20. (a) $\frac{1}{2}+\frac{2}{3}=\frac{7}{6}$

(b) $\frac{3}{7}+\left(\frac{-5}{14}\right)=\frac{6-5}{14}=\frac{1}{14}$

(c) $\frac{8}{9}+\frac{1}{3}=\frac{8+3}{9}=\frac{11}{9}$

(d) $\frac{10}{9}+\frac{25}{18}=\frac{20+25}{18}=\frac{45}{18}=\frac{5}{2}$

Answer) **a-q, b-r, c-s, d-p**

21. $\frac{k}{5}+\frac{3}{7}-\frac{1}{2}=\frac{51}{70}\longrightarrow\frac{k}{5}=\frac{51}{70}-\left(\frac{6-7}{14}\right)=\frac{57}{70}+\frac{1}{14}\Rightarrow\frac{k}{5}=\frac{51+5}{70}=\frac{56}{70}\longrightarrow k=\frac{56}{70}\times 5=\frac{56}{14}=4$

Answer) **4**

22. $\frac{a}{b}+\frac{a}{c}+\frac{a}{d}=\frac{1000}{5}+\frac{1000}{25}+\frac{1000}{125}$

Answer) **B,C**

23. $2-\frac{3}{2}+1-0-1+5=7-\frac{3}{2}=\frac{14-3}{2}=\frac{11}{2}$

Answer) **B,C**

24. $\frac{5+8-7}{2}-6=3-6=-3$

Answer) **B,C**

25. $A=\frac{5}{6}-\left(\frac{5}{6}\right)=0$

Answer) **A**

26. $B=\frac{7}{4}-\left(\frac{3}{4}\right)=\frac{4}{4}=1$

Answer) **3**

27. $B-A=1-0=1$

Answer) **A**

Worksheet – 3
Level – I

1) **D** 2) **C** 3) **C** 4) **B** 5) **A** 6) **C** 7) **B**

8) **C** 9) **B** 10) **A**

Level – II

1. Let the multiplied number is ' x ' $\Rightarrow 6\frac{2}{9}\times x=4\frac{4}{9}\Rightarrow\frac{56}{9}\times x=\frac{40}{9}\Rightarrow x=\frac{40}{56}\Rightarrow x=\frac{5}{7}$

Answer) **B**

2. $-(-x)=+x$

Answer) **B**

3. $\frac{2}{5}\times\frac{-4}{9}=\frac{-8}{45}$; The reciprocal of $\frac{-8}{45}$ is $\frac{-45}{8}$

Answer) **D**

4. $\frac{2}{5} \times 5\frac{1}{5} = \frac{2}{5} \times \frac{26}{5} = \frac{52}{25}$

   Answer) **C**

5. $-3\frac{1}{5} \times \frac{3}{8} \times 3\frac{1}{3} = \frac{-16}{5} \times \frac{3}{8} \times \frac{10}{3} = \frac{-16 \times 2}{8} = -4$

   Answer) **D**

6. $\frac{12}{11}$

   Answer) **A**

7. $\frac{5}{129} \times \frac{129}{5} = 1$

   Answer) **C**

8. $\frac{8}{21} \times \frac{21}{8} = 1$

   Answer) **C**

9. Let the cost per 1 kg = x/−$\Rightarrow 6\frac{1}{4} \times x = R\ \text{s}.400$

   $\Rightarrow \frac{25}{4} \times x = 400 \Rightarrow x = 400 \times \frac{4}{25}$

   $x = 16 \times 4 \Rightarrow x = 64/-$

   Answer) **B**

10. $[(48) \div 16] \div 3 = (3) \div 3 = 1$

    Answer) **A**

11. $(-6 + 30) \div (14 - 2) = 24 \div 12 = 2$

    Answer) **B**

12. Answer) **a-q; b-s; c-p; d-r**
13. Answer) **B,D**
14. Answer) **C**
15. Answer) **C**
16. Answer) **B**
17. Answer) **D**
18. Answer) **D**
19. Answer) **a-r; b-s; c-p; d-q**
20. Answer) **A**

Worksheet – 4

Level – I

1) **A** 2) **C** 3) **D** 4) **B** 5) **C** 6) **A** 7) **C**

8) **A** 9) **B** 10) **C** 11) **B**

Level – II

1) **C** 2) **D** 3) **C** 4) **C** 5) **B** 6) **2** 7) **D**

8) **D** 9) **C** 10) **A** 11) **6** 12) **7** 13) **A,B** 14) **A**

15) **B** 16) **A** 17) **C** 18) **a-p,r,s,t; b-s,t; c-p,r; d-q** 19) **48**

20) **57** 21) **1,2**

22. $P + Q = \frac{1}{3} + \frac{3}{11} = \frac{11+9}{33} = \frac{20}{33} = 0.6060$ ...

Answer) **B**

23. $P - Q = \frac{1}{3} - \frac{3}{11} = \frac{11-9}{33} = \frac{2}{33} = 0.06060$ ....

Answer) **B**

24. $(P + Q)(P - Q) = \frac{20}{33} + \frac{2}{33} = \frac{22}{33} = \frac{2}{3} = 0.66$

Answer) **B**

Worksheet – 5

Level – I

1) **B** 2) **D** 3) **C** 4) **C** 5) **C** 6) **A** 7) **D**
8) **D** 9) **D** 10) **C**

Level – II

1. $\frac{-48}{60} = \frac{12\times-4}{12\times5} = \frac{-4}{5}$

Answer) **C**

2. one number is $\frac{-12}{3}$, and let 2nd number be ' x '

Sum of two numbers = $\frac{-1}{3}$

$\frac{-12}{3} + x = \frac{-1}{3} \Rightarrow x = \frac{12}{3} - \frac{1}{3} = \frac{11}{3}$

Answer) **A**

3. Answer) **B**

4. Total weight of fruits are $\Rightarrow 3\frac{1}{2} + 4\frac{3}{4} \Rightarrow \frac{7}{2} + \frac{19}{4} \Rightarrow \frac{14+19}{4} \Rightarrow \frac{33}{4} = 8\frac{1}{4}$

Answer) **C**

5. $35 \times \frac{2}{7} = 10 \Rightarrow 35 - 10 = 25$

Answer) **25**

6. 6 Bowls cost = Rs.96 ; 1 Bowl cost = 96/6 = Rs. 16

Answer) **C**

7. 1 Rupee = 100 paise then 75 paise = 75/100 = 0.75

Answer) **B**

8. $5 + x = 9 \Rightarrow x = 9 - 5 = 4 \ \therefore x = 4$

Answer) **D**

9. $20 \times x = 60 \Rightarrow x = \frac{60}{20} = 3 \ \therefore x = 3$

Answer) **A**

10. Let the number be x ∴ by given problem

    $5x - 7 = 63 \Rightarrow 5x = 70 \Rightarrow x = 14$

    Answer) 14

11. Answer) **B**

12. $x \div 9 = 4 \Rightarrow x = 4 \times 9 = 36 \ \therefore x = 36$

    Answer) **A**

13. 13 $\left(\frac{-1}{3}\right)^3 \times \left(\frac{3}{2}\right)^2 = \frac{-1}{27} \times \frac{9}{4} = \frac{-1}{12}$

    Answer) **D**

14. $x \times y = 15 \Rightarrow x \times (-10) = 15 \Rightarrow x = \frac{15}{-10} = \frac{-3}{2} \ \therefore x = \frac{-3}{2}$

    Answer) **C**

15. $\frac{-4}{5} = \frac{-x}{y} \Rightarrow x + y = 9$

    Answer) **9**

16. $\frac{2}{3} = \frac{2\times16}{3\times16} = \frac{32}{48} \Rightarrow \frac{5}{6} = \frac{5}{6} \times \frac{8}{8} = \frac{40}{48}$

    Answer) **A,C**

17. Answer) **D**

18. Answer) **A**

19. Answer) **72**

20. Answer) **B,D**

21. Answer) **A,C,D**

4. SQUARE ROOTS AND CUBE ROOTS

Worksheet – 1

Level – I

1. $64 = 8 \times 8 \Rightarrow \sqrt{64} = 8$

   Answer) **C**
2. $121 = 11 \times 11 \Rightarrow \sqrt{121} = 11$

   Answer) **C**
3. $1024 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$ (By prime factorization)

   $\sqrt{1024} = 2 \times 2 \times 2 \times 2 \times 2 = 32$

   Answer) **A**
4. $\sqrt{\frac{25}{9}} = \sqrt{\frac{5\times5}{3\times3}} = \frac{5}{3}$

   Answer) **C**
5. $\sqrt{\frac{144}{169}} = \frac{\sqrt{144}}{\sqrt{169}} = \frac{12}{13}$

   Answer) **A**
6. $243 = \underline{3 \times 3} \times \underline{3 \times 3} \times 3$ (By prime factorization) multiply with 3 to make perfect square

   Answer) **C**
7. $192 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3$ (By prime factorization)multiply with 3

   Answer) **C**
8. Conceptual

   Answer) **C**
9. Conceptual

   Answer) **B**
10. Conceptual

    Answer) **B**

Level – II

1. $(64)^2 = 4096$

   Answer) **B**
2. 36 (last digit is even number, sum of digits is divisible by 3 and 36 is divisible by 4)

   Answer) **A**
3. Find the square root by long division method and then the remaining no.should be subtracted from the 2509

   Answer) **B**
4. $\frac{4}{7} = \frac{n}{49} \Rightarrow n = 28$

   Answer) **D**

5. $\sqrt{\frac{25}{100}} = \frac{5}{10} = 0.5$

   Answer) **A**

6. $\sqrt{11 \times 11 \times 11 \times 11} = 11 \times 11 = 121$

   Answer) **121**

7. $\sqrt{\frac{441}{4}} = \frac{21}{2}$

   Answer) **C**

8. $\sqrt{\frac{\frac{289}{1000}}{\frac{121}{100000}}} = \sqrt{\frac{289}{1000} \times \frac{100000}{121}} = \frac{17 \times 10}{11} = \frac{170}{11}$

   Answer) **A**

9. The smallest four digit number is 1000 , which is not a perfect square as $\sqrt{1000} = 31.622$. Hence the number after 31.622 , that is 32 's square will be four digit perfect square. $32 \times 32 = 1024$.

   Answer) **C**

10. $\sqrt{\frac{\frac{361}{1000}}{\frac{169}{100000}}} = \sqrt{\frac{361}{1000} \times \frac{100000}{169}} = \frac{190}{13}$

    Answer) **D**

11. $x^2 = 40 \times 90 \Rightarrow x = \sqrt{40 \times 90} = \sqrt{3600} = 60$

    Answer) **6**

12. $\frac{14}{\sqrt{x}} \cdot \frac{30}{\sqrt{x}} = 4 \Rightarrow x = 105$ 13. $16 \times 16 = x \Rightarrow x = 256$

    Answer) **C**

13. $\frac{9^2}{9} = 9$

    Answer) **A**

14. $128 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$

    2 must be multiplied to 128 to make it a perfect square.

    Answer) **B**

15. Answer) **A**

16. $\sqrt{\frac{25-9}{81}} = \sqrt{\frac{16}{81}} = \frac{4}{9}$

    Answer) **B**

17. $\sqrt{41 - \sqrt{21 + \sqrt{19 - 3}}} = \sqrt{41 - \sqrt{21 + 4}} = \sqrt{41 - 5} = \sqrt{36} = 6$

    Answer) **C**

18. Nearest perfect square to 84 is 81.

Answer) **B**

19. $\sqrt{225} = 15 \Rightarrow$ number of digits $= 2$.

Answer) **A**

20. $\sqrt{(272+128)(272-128)} = \sqrt{400 \times 144} = \sqrt{(20 \times 12)^2} = 240$

Answer) **240**

21. $\sqrt{0.01} + \sqrt{0.81} + \sqrt{1.21} + \sqrt{0.009} = 0.1 + 0.9 + 1.1 + 0.03 = 2.13$

Answer) **D**

22. $\frac{\sqrt{24}+\sqrt{216}}{\sqrt{96}} = \frac{\sqrt{6\times4}+\sqrt{36\times6}}{\sqrt{16\times6}} = \frac{2\sqrt{6}++6\sqrt{6}}{4\sqrt{6}} = \frac{8\sqrt{6}}{4\sqrt{6}} = 2$

Answer) **B**

23. $3\sqrt{5} + \sqrt{125} = 17.88 \Rightarrow 3\sqrt{5} + 5\sqrt{5} = 17.88 \Rightarrow 8\sqrt{5} = 17.88 \Rightarrow \sqrt{5} = 2.235$
Now $\sqrt{80} + 6\sqrt{2} = \sqrt{16 \times 5} + 6\sqrt{2} = 4\sqrt{5} + 6\sqrt{2} = 4(2.235) + 6(1.414) = 17.424$

Answer) **A**

24. 24. $\sqrt{8} + 2\sqrt{32} - 3\sqrt{128} + \sqrt{50}$

$\sqrt{4 \times 2} + 2\sqrt{16 \times 2} - 3\sqrt{64 \times 2} + \sqrt{25 \times 2}$

$2\sqrt{2} + 2 \times 4\sqrt{2} - 3 \times 8\sqrt{2} + 5\sqrt{2} \Rightarrow 2\sqrt{2} + 8\sqrt{2} - 24\sqrt{2} + 5\sqrt{2}$

$15\sqrt{2} - 24\sqrt{2} = -9\sqrt{2} = -9(1.414) = -12.726$

Answer) **B**

25. Conceptual

Answer) **a-r;b-s;c-t;d-p**

26. $\left(5 - \sqrt{y^2 - 25}\right)^2 = 25 + y^2 - 25 - 10\sqrt{y^2 - 25}$

$= y^2 - 10\sqrt{y^2 - 25}$ (or) $= y^2 - \sqrt{(10y)^2 - 10^2 \times 5^2} = y^2 - \sqrt{(10y)^2 - (50)^2}$

Answer) **16**

27. Conceptual.

Anwer) **C**

28. $\left(5 - \sqrt{y^2 - 25}\right)^2 = 25 + y^2 - 25 - 10\sqrt{y^2 - 25}$

$= y^2 - 10\sqrt{y^2 - 25}$ (or) $= y^2 - \sqrt{(10y)^2 - 10^2 \times 5^2} = y^2 - \sqrt{(10y)^2 - (50)^2}$

Answer) **C,D**

29. Conceptual

Anwer) **C**

30. Conceptual

Answer) **B**

31. Conceptual

Answer) **B**

32. Conceptual

Answer) **a-r;b-p;c-q;d-t**

33. Conceptual

Answer) **A**

34. Conceptual

Answer) **a-s;b-p;c-t;d-q**

35. $\frac{32.4}{x} = 4 \Rightarrow x = \frac{32.4}{4} \Rightarrow x = 8.1$

Answer) **8.1**

TOPIC – 2

CUBE AND CUBE ROOTS

Worksheet – 2

Level – I

1) **B** 2) **A** 3) **A** 4) **D** 5) **B** 6) **C** 7) **C**

8) **D** 9) **C** 10) **D**

Level – II

1. 3/4

Answer) **A**

2. 5

Answer) **D**

3. 50

Answrer) **C**

4. $13 \times 13 \times 13 = 2197$

Answrer) **C**

5. $\sqrt[3]{27000} = \sqrt[3]{2^3 \times 3^3 \times 5^3} = 30$

Answrer) **30**

6. $(3^3 + 4^3 + 5^3)^{\frac{1}{3}} = (27 + 64 + 125)^{\frac{1}{3}} = (216)^{\frac{1}{3}} = (6^3)^{\frac{1}{3}} = 6$

Answrer) **A**

7. $6^3 - 5^3 = 216 - 125 = 91$

Answrer) **A**

8. Answer) **D**

9. $\sqrt[3]{\frac{343}{125}} \times \sqrt[3]{32 \times 6} = \frac{7}{5} \times \sqrt[3]{64 \times 3} = \frac{7}{5} \times 4 \times \sqrt[3]{3} = \frac{28}{5} \times \sqrt[3]{3}$

Answrer) **D**

10. Answer) **B**

11. $\sqrt[3]{4\frac{508}{1331}} = \sqrt[3]{\frac{5832}{1331}} = \sqrt[3]{\frac{2^3\times3^3\times3^3}{11^3}} = \frac{2\times3\times3}{11} = \frac{18}{11} = 1\frac{7}{11}$

Answrer) **C**

12. $\sqrt[3]{\sqrt{0.000064}} = \sqrt[3]{0.008} = \sqrt{(0.2)^3} = 0.2$

Answrer) **B**

13. Conceptual

Answrer) **6**

14. $\sqrt[3]{16\frac{16}{27}} = \frac{4}{3}\times\sqrt[3]{x} \Rightarrow \sqrt[3]{\frac{448}{27}} = \frac{4}{3}\times\sqrt[3]{x} \Rightarrow \sqrt[3]{\frac{2^3\times2^3\times7}{33}} = \frac{4}{3}\times\sqrt[3]{x}$

$\frac{2\times2\times\sqrt[3]{7}}{3} = \frac{4}{3}\times\sqrt[3]{x} \Rightarrow \sqrt[3]{7} = \sqrt[3]{x} \Rightarrow x = 7$

Answrer) **A**

15. $125\sqrt[3]{a^6} - \sqrt[3]{125a^6} = 125a^2 - (5^3a^2)\frac{1}{3} = 125a^2 - 5^3 = 120a^2$

Answrer) **A**

16. Conceptual.

Answrer) **C**

17. $\left(\sqrt{15^2+18^2}\right)^3 = (\sqrt{225+324})^3 = (\sqrt{549})^3 = (23)^3 = 12{,}167$

Answrer) **C**

18. $3\sqrt{448}\times3\sqrt{392} = 3\sqrt{2^3\times2^3\times7}\times3\sqrt{2^3\times7^2} = 3\sqrt{2^3\times2^3\times2^3\times7^3} = 2\times2\times2\times7 = 56$

Answrer) **54**

19. $27\times(-4) = -28$

Answrer) **B**

20. Ratio $= 2:3:4 \Rightarrow$ Sum of their Cubes $= 33957$

$(2x)^2 + (3x)^3 + (4x)^3 = 33957 \Rightarrow 8x^3 + 27x^364x^3 = 33957$
$99x^3 = 33957 \Rightarrow x^3 = 343 \Rightarrow x = 7$

Answrer) **B**

21. $\frac{3125}{25} / \frac{25}{\sqrt[3]{125}} = \frac{3125}{25}\times\frac{\sqrt[3]{125}}{25} \Rightarrow 125\times\frac{5}{25} = 5\times5 = 25$

Answrer) **C**

22. $\sqrt[3]{\frac{x}{11664}} = \frac{2}{3} \Rightarrow \frac{\sqrt[3]{x}}{\sqrt[3]{2\times2^3\times3^3\times3^3}} = \frac{2}{3} \Rightarrow \frac{\sqrt[3]{x}}{2\times3\times3\times\sqrt[3]{2}} = \frac{2}{3}$

$\frac{\sqrt[3]{x}}{6\sqrt[3]{2}} = 2 \Rightarrow \frac{x}{6^3\times2} = 2^3 \Rightarrow x = 3456$

Answrer) **3456**

23. Conceptual.

    Answrer) **a-s; b-p; c-q;d-t;e-r;f-q**

24. Conceptual.

    Answrer) **A**

25. Conceptual.

    Answrer) **B,C,D**

26. $\sqrt[3]{0.000110592} = \sqrt[3]{\frac{110592}{1000000000}} = \sqrt[3]{\frac{2^3\times2^3\times12^3}{10^3}} = \frac{2\times2\times12}{10^3} = \frac{48}{1000} = 0.048$

    Answrer) **0.048**

27. $\sqrt[3]{\frac{0.027}{0.008}} \div \sqrt{\frac{0.09}{0.04}} - 1 = \frac{0.3}{0.2} \div \frac{0.3}{0.2} - 1 = \frac{0.3}{0.2} \times \frac{0.2}{0.3} - 1 = 1 - 1 = 0$

    Answrer) **A**

28. $\sqrt[3]{125 \times 64} = 3\sqrt{5^3 \times 4^3} = 20$

    Answrer) **C**

29. $\sqrt[3]{\frac{0.216}{27}} = \sqrt[3]{\frac{(0.6)^3}{3^3}} = \frac{0.6}{3} = 0.2$

    Answrer) **A**

30. Conceptual.

    Answrer) **a-s; b-t; c-q; d-p; e-r**

31. Conceptual.

    Answrer) **B**

32. Conceptual.

    Answrer) **A,B**

33. Conceptual.

    Answrer) **A,B,C**

34. $\sqrt[3]{27} + \sqrt[3]{0.008} + \sqrt[3]{0.064} = 3 + 0.2 + 0.4 = 3.6$

    Answrer) **3.6**

TOPIC – 3

PROPERTIES OF NUMBERS

Worksheet – 3

Level – I

1) **A** 2) **C** 3) **B** 4) **B** 5) **B** 6) **A** 7) **B**

8) **C** 9) **A** 10) **C**

Level – II

1) **B**	2) **B**	3) **D**	4) **A**	5) **1**	6) **B**	7) **C**
8) **C**	9) **C**	10) **C**	11) **B**	12) **C**	13) 5	14) **C**
15) **A**	16) 8	17) **B**	18) **D**	19) **B**	20) 0	21) **A,B**
22) **A**	23) **D**	24) **B**	25) **D**	26) **a-q;b-s;c-p;d-r**		27) **A**
28) **a-s;b-p;c-r;d-q**		29) **C**	30) **B**	31) **D**	32) **A**	33) **A,B**
34) **D**	35) **a-p;b-r;c-s;d-q**					

6. Sum of additive inverse of first five natural numbers is $-1-2-3-4-5=-15$

7. Product of additive inverse of first five whole numbers is 0

9. Additive inverse of $2\frac{3}{4}$ is $\frac{-11}{4}$

14. $xy=1 \Rightarrow y=\frac{1}{x} \Rightarrow x=-4\frac{1}{5}=-\frac{21}{5} \Rightarrow y=\frac{-5}{21}$

15. $\frac{7}{9}\times\frac{243}{343}=\frac{27}{49}$

17. Additive inverse of $-12+16-2$ is -2

    Product of $-2,-3,-4$ is -24

    Sum $=-2-24=-26$

18. $\frac{2}{11}\times\frac{14}{-5}=\frac{28}{-55}$

29. $\frac{-2}{3}+\frac{5}{7}=\frac{-14+15}{21}=\frac{1}{21}$

30. $\frac{5}{7}+\left(\frac{-1}{6}\right)=\frac{30-7}{42}=\frac{23}{42}$

31. $\frac{1}{21}+\left(-\frac{1}{6}\right)=\frac{2-7}{42}=\frac{-5}{42}$

5. COMMERCIAL MATHEMATICS

BUSINES MATHMATICS

Worksheet – 1

Level – I

1) D 2) C 3) C 4) A 5) B 6) B 7) B

8) B 9) B 10) C

1. $4,9,12,\text{x} \Rightarrow 4 \times x = 9 \times 12 \Rightarrow x = \frac{9\times12}{4} = 27$

2. $\frac{1}{5}:\frac{1}{x} = \frac{1}{x}:\frac{1}{25}$

$$\left(\frac{1}{x}\right)^2 = \frac{1}{5}\times\frac{1}{125}$$
$$\left(\frac{1}{x}\right)^2 = \frac{1}{625}$$
$$\frac{1}{x} = \frac{1}{25}$$
$$x = 25$$

$x^2 = 625 \Rightarrow x = 25$

3. $x:y = 5:2 \Rightarrow \frac{x}{y} = \frac{5}{2} \Rightarrow \frac{8x+9y}{8x+2y} = \frac{8(5)+9(2)}{8(5)+2(2)} = \frac{40+18}{40+4} = \frac{58}{44} = 29:22$

4. Duplicate ratio of $2:3$ is $= 2^2:3^2 = 4:9$

5. $\frac{25 \text{ paise}}{8 \text{ rupees}} = \frac{25}{8\times100} = \frac{25}{800} = 1:32$

6. Sub duplicate ratio of $121:144$ is $11:12$

7. 15% of x $= 20\%$ of y $\Rightarrow \frac{15}{100}\times x = \frac{20}{100}\times y \Rightarrow 15x = 20y \Rightarrow \frac{x}{y} = \frac{20}{15} = x:y = 4:3$

8. Ratio of $4^{3.5}:2^5 \Rightarrow (2^2)^{3.5}:2^5 \Rightarrow 2^7:2^5 \Rightarrow \frac{2^7}{2^5} = \frac{2^2}{1} = 4:1$

9. The ratio of 20 cm and 3 m $= \frac{20 \text{ cm}}{3 \text{ m}} = \frac{20 \text{ cm}}{3\times100} = \frac{20}{3\times100} = \frac{1}{15} = 1:15$

10. Ratio of three numbers $= 3:4:7$

$$3x \times 4x \times 7x = 18144 \Rightarrow 84x^3 = 18144 \Rightarrow x^3 = \frac{18144}{84} \Rightarrow x^3 = 216 \Rightarrow x = 6$$

$$No's = 3 \times x = 3 \times 6 = 18 \Rightarrow 4 \times 6 = 24 \Rightarrow 7 \times 6 = 42$$

Level – II

1) B 2) C 3) A 4) D 5) A 6) A 7) B

8) B 9) B 10) A 11) C 12) D 13) B 14) B

15) D 16) C 17) C 18) C 19) D

1. $\frac{4}{15}A = \frac{2}{5}B \Rightarrow \frac{A}{B} = \frac{2}{5}\times\frac{5}{4} \Rightarrow \text{A:B} = 3:2 \Rightarrow \text{B share} = \frac{2}{5}\times 1210 = 2 \times 242 = 484$

2. Let the shares A, B, C and D be Rs $5x, 2x, 4x$, and $3x$, respectively then $4x - 3x = 1000 \Rightarrow x = 1000$

   B share = Rs 2x = 2x1000 = 2,000

3. Let originally the number of seats maths, physics, and biology be $5x, 7x$, and $8x$ respectively. Number of increased seats are (140% of $5x$)(150% of $7x$) and (175% of 8x) $\Rightarrow 7x, \frac{21x}{2}, 14x \Rightarrow$ the required ratio $7x : \frac{21x}{2} : 14x \Rightarrow 14x = 21x : 28x \Rightarrow 2 : 3 : 4$

4. Quantity of milk = $60 \times \frac{2}{3}$ liters = 40 ltr

   Quantity of water in it = $60 - 40 = 20$ liters $\Rightarrow$ New ratio = $1 : 2$

   let the quantity of water be added further be x liters

   then milk : water = $\frac{40}{20+x} \Rightarrow$ now $20 + x = 80 \Rightarrow x = 60$

   Quantity of water to be added = 60 litre

5. $\frac{A}{3} = \frac{B}{4} = \frac{C}{5} = k \Rightarrow A : B : C = 3k : 4k : 5k = 3 : 4 : 5$

6. $A : B = 8 : 15 \Rightarrow B : C = 5 : 8 \Rightarrow A : B : C = 40 : 75 : 120 = 8 : 15 : 24$

7. The fourth proportion of 5,8,15, x $\Rightarrow 5 \times x = 8 \times 15 \Rightarrow x = \frac{8 \times 15}{5} = 24$

8. Given ratio = $3 : 4 \Rightarrow \frac{3}{4} = \frac{3 \times 4}{4 \times 4} = \frac{12}{16}$ consequent = 16

9. Let the required number be $3x$ and $4x$, then their

   LCM is $12x; 12x = 180 \Rightarrow x = \frac{180}{12} = 15 \Rightarrow$ Hence the 1st No $= 3 \times 15 = 45$

10. Let the required quantity of copper be xkg

    then $9 : 4 : : 24 : x \Leftrightarrow 9x = 4 \times 24 \Rightarrow x = \frac{4 \times 24}{9} = \frac{32}{3} \Rightarrow x = 10\frac{2}{3}$ kg

    Hence the required quantity copper = $10\frac{2}{3}$ kg

11. Originally let the number of boys and girls in the college be $7x, 8x$ respectivley their increased number is (120% of $7x$) and 110% of $8x$

    $\frac{120}{100} \times 7x$ and $\frac{110}{100} \times 8x = \frac{42x}{5}$ and $\frac{44x}{5} \Rightarrow$ required ratio $= \frac{42}{5} : \frac{44x}{5} = 21 : 22$

12. Let the original salaries of Ravi's and Sumit be

    Rs $2x$ and Rs $3x$ respectively $\Rightarrow$ then $\frac{2x+4000}{3x+4000} = \frac{40}{57}$

    $57(2x + 4000) = 40(3x + 4000) \Rightarrow 6x = 68{,}000 \Rightarrow 3x = 34{,}000$

    Sumit present salary $= 3x + 4000 = 34000 + 4000 = 38{,}000$

13. $X \times 5 = 0.75 \times 8 \Rightarrow x = 1.2$

14. Let the three No's be A,B,C

    then $A : B = 2 : 3$, and $B : C = 5 : 8 = \left(5 \times \frac{3}{5}\right) : \left(8 \times \frac{3}{5}\right) = 3 : \frac{24}{5}$

A:B:C $2:3:\frac{24}{5} = 10:15:24 \Rightarrow B = 98 \times \frac{15}{48} = 30$

15. Given ratio $= \frac{1}{2}:\frac{2}{3}:\frac{3}{4} = 6:8:9 \Rightarrow 1^{st}$ part $= \text{Rs}\left(782 \times \frac{6}{23}\right) = Rs204$

16. Let $a = 2k, b = 3k, c = 5k$

a's new salary $= \frac{115}{100}$ of $2k = \frac{115}{100} \times 2k = \frac{23k}{10}$

b's new salary $= \frac{110}{100}$ of $3k = \frac{110}{100} \times 3k = \frac{33k}{10}$

c's new salary $= \frac{120}{100}$ of $5k = \frac{120}{100} \times 5k = 6k$

New ratio $\frac{23k}{10}:\frac{33k}{10}6k = 23:33:60$

17. Let 40% of A $= \frac{2}{3}$ B $\Rightarrow$ Then $\frac{40A}{100} = \frac{2B}{3} \Rightarrow \frac{2A}{5} = \frac{2B}{3} \Rightarrow \frac{A}{B} = \frac{2}{3} \times \frac{5}{2} \Rightarrow \frac{A}{B} = \frac{5}{3} \Rightarrow A:B = 5:3$

18. Let the number of 25P, 10P and 5P Coins be x, 2x, 3x respectively

Their sum of their values $= \frac{25x}{100} + \frac{10 \times 2x}{100} + \frac{5 \times 3x}{100} = \text{Rs}\frac{60x}{100}$

$\Rightarrow \frac{60x}{100} = 30 \Rightarrow x = \frac{30 \times 100}{60} = 50 \Rightarrow$ Hence the number of 5p Coins $= 3 \times 50 = 150$

19. Suppose the quantity of alcohol and water $= 5x$ and $3x$

$\frac{4x}{3x+5} = \frac{4}{5}$ or $20x = 4(3x + 5) \Rightarrow 20x = 12x + 20 \Rightarrow 20x - 12x = 20 \Rightarrow 8x = 20 \Rightarrow x = 2.5$

Hene the quantity of alcohol $= 4 \times 2.5 = 10L$

Level – III

20) B 21) A 22) A 23) A 24) A,B,C,D 25)240 26) 25

27) B 28) D 29) 15 30) C 31) D 32) B 33) B

34) C 35) a-s; b-r; c-q;d-p

20. Ratio of 5 Rupees and 200 paise $= \frac{500}{200} = 5:2$

21. Let the third proportional to 0.36 and 0.48 be x

Then 0.36:0.48::0.48: x $\Rightarrow x = \frac{0.48 \times 0.48}{0.36} \Rightarrow x = 0.64$

22. The duplicate ratio of 6:7 is $= 6^2:7^2 = 36:49$

23. Triplicate rato of 12:7 is $12^3:7^{30} = 1728:343$

24. $\frac{6}{15} = \frac{6 \times 2}{15 \times 2} = \frac{12}{30} = 12:30 \Rightarrow \frac{6}{15} = \frac{6 \times 3}{15 \times 3} = \frac{18}{45} = 18:45$

$$\frac{6}{15} = \frac{6 \times 4}{15 \times 4} = \frac{24}{60} = 24:60 \Rightarrow \frac{6}{15} = \frac{6 \times 5}{15 \times 5} = \frac{30}{75} = 30:75$$

25. We have cost of 30 m of cloth = Rs. 450

cost of 1 m of cloth = Rs. $\frac{450}{30} = 15$

Then cost 16 m of cloth Rs. $(15 \times 16) = Rs.240$

26. 50% of $(x-y)$ = 30% of $(x+y)$

$\frac{50}{100}(x-y)=\frac{30}{100}(x+y) \Rightarrow 5(x-y)=3(x+y) \Rightarrow 2x=8y \Rightarrow 2x=8y \Rightarrow x=4y$

Required percentage $=\left(\frac{y}{x}\times 100\right)\% \Rightarrow \left(\frac{y}{4y}\times 100\right)=25\%$

27. Ratio of sides $=\frac{1}{2}:\frac{1}{3}:\frac{1}{4}=6:4:3 \Rightarrow$ largest side $=\frac{6}{13}\times 104=48$ cm

28. Let the number of boys and girls be 8x and 5x respectively. Then $5x = 160 \Rightarrow x = 32$ The total number of students $= 13x = 13 \times 32 = 416$

29. Total age of 3 boys $= 25 \times 3 = 75$ years $\Rightarrow$ Ratio of their ages $= 3:5:7$ age of the youngest $= 75 \times \frac{3}{15} = 15$ years

30. Let the prices of scooter and TV set be 7x and 5x respectively $7x-5x=8000 \Rightarrow 2x=8000 \Rightarrow x=4000 \Rightarrow$ Price of a TV Set is $= 7 \times 4000 = 28{,}000$

31. G = 19w and C = 9w

Let 1gm of gold be mixed with x gm of copper to get (1 + x)gm of the alloy 1gm gold +x gm copper = (x + 1) gm of a alloy

$19w + 9wx = (x+1) \times 15w \Rightarrow 19 + 9x = 15(x+1) \Rightarrow 15x - 9x = 19 - 15 \Rightarrow 6x = 4$

$x=\frac{4}{6}=\frac{2}{3} \Rightarrow x = 2/3 \Rightarrow$ Ratio of gold with copper $= 1:\frac{2}{3}=3:2$

32. $5^2:6^2=25:36$

33. 13:12

34.3

35. a – s; b – r, c – p; d – q

Worksheet – 2

Level – I

1) D 2) A 3) C 4) A 5) A 6) C 7) A

8) A 9) B 10) C

1. $35\%=\frac{35}{100}=\frac{7}{20}$
2. $0.25\%=\frac{25}{100\times 100}=\frac{1}{400}$
3. $\frac{3}{5}\times 100=60$
4. $22\frac{1}{2}\%=\frac{45}{2}\%=\frac{45}{2\times 100}=\frac{9}{40}$
5. $20\%\times 200=\frac{20}{100}\times 200=40$
6. $\frac{25}{2\times 100}\times 400=50$

7. $25\% \times x = 50 \Rightarrow \frac{50\times100}{25} = 200$

8. $S.I = \frac{P\times T\times R}{100} = \frac{500\times1\times10}{100} = 50 \Rightarrow$ Amount $= P + I = 500 + 50 = 550$

9. $S.I = \frac{1000\times2\times5}{100} = 100$

10. $P = \frac{100\times S.I}{T\times R} = \frac{100\times1000}{5\times2} = 10000$

Level – II

1) B	2) C	3) A	4) D	5) B	6) C	7) C
8) C	9) A	10) C	11) B	12) A	13) A	14) A
15) B	16) B	17) C	18) C			

1. $\frac{5}{100} = \frac{1}{20}$

2. Original price = 15,000/−, Decreased price = 10,000 decrease = 15000

   Decrease % $= \left(\frac{5000}{15000}\times100\right)\% = 33\frac{1}{3}\%$

3. Increased % $= 12\frac{1}{2}\% = 25/4 \times 100 = 1/8$ ∴ Increased value $= 400 \times 1/8 = 50$

4. $A = 5000\left(1 + \frac{\frac{7}{4}\times18}{100}\right) = 6575/-$ 5. $A = 7500\left(1 + \frac{\frac{9}{4}\times16}{100}\right) = 10200/-$

5. $S.I = \frac{PTR}{100} = 4800 \times 6\frac{1}{4} \times \frac{4}{100} = 4800 \times \frac{25}{4} \times \frac{4}{100} = 1200/-$

6. The monthly consumption be ' x ' liters ∴ 25% of $x = \frac{25x}{100} = \frac{x}{4}$ litres

   $\Rightarrow$ 20litres $= \frac{x}{4}$ litres $\Rightarrow x = 80$

7. Let the amount of original bill = Rs. x, given 8% of $x = \frac{8x}{100} = \frac{2x}{25}$

   Given $x - \frac{2x}{25} = 230 \Rightarrow x\left(\frac{23}{25}\right) = 230 \Rightarrow x = 250/-$

9. Time = T = 16 days in Feb + 31 days in March +30 days in April +31 days in May +30 in June +31 in July +31 in August +30 in September = 230 days

   $R = 24\%, P = 73000/-, T = \frac{230}{365}$ years

   S.I. $= \frac{PTR}{100} = \frac{73000\times24}{100} \times \frac{230}{365} = 240 \times 46 = 11040/-$

10. marks scored by david = 450, Total marks = 600 percentage of marks $= \left(\frac{450}{600}\times100\right)\% = 75\%$

11. $\frac{1584}{1440} = \left(1 + \frac{\frac{10}{3}\times R}{100}\right) \Rightarrow \frac{11}{10} = 1 + \frac{10R}{300} \Rightarrow R = 3$ ∴ $A = 920 \times \left(1 + \frac{5\times3}{100}\right) = 1058$

12. S.I : $\frac{P\times5\times6.25}{100} = \frac{8\times4\times\frac{5}{2}}{100\ s} \Rightarrow P = \frac{36\times100}{6.25} = \frac{36\times100\times100}{625} = 576/-$

13. Raju's present salary = 4200/−,

    increase % of raju's salary = 12(1/2)% = 25/2% new 12(1/2) of 4200

    $= \frac{\frac{25}{2}\times 4200}{100} = 525 \Rightarrow$ Raju's next year salary = 4500 + 525 = 4725/−

14. Present cost of house = 6000/−15% of 6000 $= \frac{15\times 60000}{100} = 9000$

    The cost of house in 1st year = 60000 + 9000 = 69000/−

    20% of 69000 $= \frac{20\times 69000}{100} = 13800 = 13800$

    The cost of house at the end of 2 nd year = 69000 −13800 = 55,200/-

15. $1260 = P\left(1+\frac{2R}{100}\right) \Rightarrow P = \frac{1260\times 50}{50+R}; 1350 = P\left(1+\frac{5R}{100}\right) \Rightarrow P = \frac{1350\times 20}{20+R}$

    $\Rightarrow \frac{1350\times 20}{20+R} = \frac{1260\times 50}{50+R} \Rightarrow R = 2.5\%$; sum $= \frac{1260\times 50\times 10}{2.5} = 1200/-$

16. Let the max. marks = x, marks got by rama = 40

    now, 40% of $x = \frac{40x}{100} = \frac{2x}{5}$ clearly , $80m + 40m = \frac{2x}{5}m \Rightarrow 120m = \frac{2x}{5} = x = 300m$

17. $A = 14600 \times \left(1+\frac{2\times 12}{100}\right) = 14600 \times \frac{31}{25} = 18104 \Rightarrow$ cost of watch = 18104 − 15304 = 2800

18. Let the first part is x; $\frac{x\times 5\times 6}{100} = \frac{(6600-x)\times 3\times 12}{100} \Rightarrow 5x = 39600 - 6x \Rightarrow x = 3600$ Let the second part = $6600 - x = 6600 - 3600 = 3000/-$

Level – III

19) A	20) A,B,D	21) A	22) C	23) D	24) **a-p;b-r;c-s;d-q**
25) 960	26) C	27) 25	28) 4	29) 1	30) 3
31)a-r;b-p;c-q;d-s		32) 4	33) 1	34) 3	35) 40000

19. Increased % $= 12\frac{1}{2}\% = 25/4 \times 100 = 1/8$ ∴ Increased value = $400 \times 1/8 = 50$

20. $80 \times \frac{20}{100} = 16$.Then the result is 80 + 16 = 96.

21. Given x + 40% of $x = 840, \Rightarrow x + \frac{40+x}{100} = 840 \Rightarrow x\left(1+\frac{4}{10}\right) = 840$

    $\Rightarrow x\left(\frac{7}{5}\right) = 840 \Rightarrow x = \frac{840\times 5}{7} \Rightarrow x = 600$

22. Increased % = income = 12500 − 1000 = 2,500

    ∴ increased % $= \left(\frac{2500}{10000}\times 100\right)\% = 25\%$

23. Increased population = (484 − 440) lakhs = 441 akhs

    ∴ Increase % $= \left(\frac{44}{440}\times 100\right)\% = 10\%$

24. a) $P = Rs1500, R = 12\%$ p.a and $T = 3\frac{1}{4}$yrs = 13/4yrs

$\therefore$ S.I $= \frac{\text{PTR}}{100} = \frac{1500\times\frac{13}{4}\times12}{100} = 585/-$

b) P $= 2500/-$, R $= 6\%$PA and T $= 5/2$yrs $\therefore$ S.I $= \frac{2500\times6\times\frac{5}{2}}{100} = 375/-$

A $=$ P $+$ S.I $= 2500 + 375 = 2875/-$

c) P $= 5000/-$, R $= 9\%$ PA & T $= 1/4$ yrs $\therefore$ S.I $= \frac{5000\times9\times\frac{1}{4}}{100} =$ Rs. $112.50/-$

d) S.I $=\therefore$ S.I $= \frac{9600\times\frac{15}{2}\times\frac{5}{12}}{100} =$ Rs. $300/-$A $= 9600 + 300 = 9900/-$

25. S.I. $= \frac{8000\times4\times3}{100} = 960/-$

26. Let P $=$ x$/-$, A $= 4$x, T $= 16$yrs $\therefore$ R $= \frac{4x}{x} = 1 + \frac{16\times R}{100} \Rightarrow 3 = \frac{4R}{25} \Rightarrow R = \frac{75}{4} = 18\frac{3}{4}\%$

27. Number of students in class $= 80$

The number of students who passed in 1st class $= 60$

The number of students who passed in 2nd class $= 80 - 60 = 20$

$\therefore$ Percentage of students who passed in 2nd class $= \left(\frac{20}{80}\times100\right)\% = 25\%$

28. Let principal be P.

$P + \frac{PTR}{100} = 6000 \Rightarrow P + \frac{P\times10\times2}{100} = 6000 \Rightarrow P + \frac{P}{5} = 6000 \Rightarrow \frac{6P}{5} = 6000$

$\Rightarrow P = 6000\times\frac{5}{6} = 5000/-$

29. Amount $= 11840/-$; Principal $= 8000/-$; Simple Interest $= 11840 - 8000 = 3840/-$

Rate $= 8\%$, T $=?$ $\therefore$ T $= \frac{100\times\text{S.I}}{\text{P}\times\text{R}} = \frac{3840\times100}{8000\times8} = 6$ years

30. Let principal be $100/-$; Amount will be $500/-$

S.I. $= 500 - 100 = 400/-$; Rate $= 25\%$, Time $=?$. T $= \frac{100\times\text{ S.I}}{P\times R} = \frac{100\times400}{100\times25} = 16$ years.

31. a) $1500\times\frac{30}{100} = 450$

b) $70\times\frac{40}{100} = 28$

c) $\frac{25}{100}\times600 = 150$

d) $\frac{26}{100}\times150 = 39$

32. Rs. $\frac{4500\times10\times\frac{1}{5}}{100} = 90$

33. P $+$ S.I $=$ Rs. $4500 + 90 =$ Rs. 4590

34. $\text{R}^1 = 2\text{R}, \text{T}^1 = 2$ T $\therefore$ S.I $\Rightarrow \frac{\text{PR}^1\ \text{T}^1}{100} = \frac{4(\text{PRT})}{100} = \text{Rs.}\,4(90) = \text{Rs.}\,360$

35. Let the population 2yrs ago $=$ x, now, 10% of x $= 10$x$/100 =$ x$/10$

The population 1yr after = x + x/10 = 11x/10

10% of 11x/10 = $\frac{10\times\frac{11x}{10}}{100} = \frac{11x}{100}$

now, the population 1yr after present pupalation = $\frac{11x}{10} + \frac{11x}{100}$

$\Rightarrow 48400 = \frac{11x}{10}\left(1+\frac{1}{10}\right) = \frac{11x}{10}\times\frac{11}{10}x = \frac{48400\times100}{11\times11} = 40000$

Worksheet – 3

Level – I

1) D 2) C 3) C 4) D 5) B 6) D 7) B

8) A 9) B 10) C

1. Gain = S.P-C.P = 500 – 450 = 50 ⇒ Gain % = $\frac{50}{450}\times100 = \frac{500}{45} = \frac{100}{9}$ (or) $11\frac{1}{9}\%$
2. Gain = S.P-C.P = 1080 – 900 = 180 Rs
3. Gain = S.P-C.P = 900 – 675 ⇒ 225 Rs ⇒ Gain percant == $\frac{225}{675}\times100 = \frac{100}{3}\% = 33\frac{1}{3}\%$
4. C.P. of radio = 1500Rs ⇒ S.P of Radio = Rs 1230 loss = C.P. - S.P = 1500 – 1230 = 270 Rs
5. Total selling = 150 mangoes. Profit = Price of 30 mangoes

   ∴ CP = total selling - profit = (150 – 30) mangoes = 120 mangoes

   ∴ gain % = $\frac{30}{120}\times100 = 25$
6. S.P of 50mts cloth = 50x. ⇒ gain = 15x.

   C.P. of 50 m cloth = 50x – 15x = 35x.

   gain percent = $\frac{\text{gain}}{C.P.}\times100 = \frac{15x}{35x}\times100 = \frac{1500}{350} = 42\frac{6}{7}\%$
7. C.P. of an article = Rs 500 ⇒ S.P. of an article = Rs 600

   SP > CP ⇒ Profit. ⇒ Profit = S P - C P. ⇒ Profit = 600 – 500 = 100 Rs
8. C.P = S.P - Profit = 2390 – 120.50 = $Rs2269.50$
9. S.P > C.P then profit will come. ⇒ Profit = 565 – 500 = 65 Rs

   Profit % = $\frac{\text{Pr ofit}}{CP}\times100 \Rightarrow \frac{65}{500}\times100 = 13\%$
10. C.P of pack = 20Rs ⇒ profit made an each packet = 4Rs

    Profit % = $\frac{\text{Pr ofit}}{CP}\times100 = \frac{4}{20}\times100 = 20\% \Rightarrow$ Profit % = 20%

Level – II

1) C 2) C 3) B 4) C 5) C 6) C 7) B

8) A 9) D 10) C 11) A 12) A 13) C 14) A

15) D 16) A 17) C 18) D 19) D 20) A,B 21) C,D

22) C 23) D 24) 1350 25) A 26) B 27) C 28) B

29) A 30)1650 31) B 32)A,B 33) B 34) 10%

35) a-r,b-s,c-p,d-q

1. CP = 250Rs S · P = 300Rs

   SP > CP = Profit will come

   Profit = S.P. - C. $P = 300 - 250 = 50$ Rs

   Profit % $= \frac{\text{Profit}}{C.P} \times 100 = \frac{50}{250} \times 100 = 20\%$

2. C.P of 600 tables $= 600 \times 80 =$ Rs 48,000

   Profit on whole = 20%

   $\therefore$ S.P. $= \frac{C.P \times (100+\text{Pr ofit \%})}{100} = \frac{48000(100+20)}{100} = \frac{48{,}000 \times 120}{100} =$ Rs 57,600

   But selling price of 500 tables of the rate = 90 Rs each $= 500 \times 90 = 45{,}000$Rs

   $\therefore$ SP of 100 tables $= 57600 - 45{,}000 = 12{,}600$Rs $\Rightarrow$ SP of 1 table $= \frac{12600}{100} = 126$Rs

3. $\therefore$ C.P $= \frac{100}{100-18} \times 1230 = 1500 \Rightarrow$ Now S.P = 1600

   Profit = S.P-C.P $= 1600 - 1500 = 100 \Rightarrow$ Profit % $= \frac{\text{profit}}{\text{C.P}} \times 100 = \frac{100}{1500} \times 100 =$

   $6\frac{2}{3}\%$

4. C.P = Rs. 1400 $\Rightarrow$ Loss % = 15%

   $\text{S.P} = \frac{100-\text{loss\%}}{100} \times \text{C.P} = \frac{100-15}{100} \times 1400 =$ Rs. 1190

5. C.P $= 20 \times 375 =$ Rs. 7500 $\Rightarrow$ S.P $= 20 \times 12 \times 33 =$ Rs. 7920

   If S.P > C.P profit will come. $\Rightarrow\therefore$ profit = S.P - C.P $= 7920 - 7500 =$ Rs. 420

   Profit % $= \frac{\text{profit}}{\text{C.P}} \times 100 = \frac{420}{7500} \times 100 = 5.6\%$

6. $\text{C.P} = (150 \times 10) + 100 = 1500 + 100 =$ Rs. 1600

   S.P $= 150 \times 12 =$ Rs. 1800 $\Rightarrow$ S.P > C.P $\Rightarrow$ Profit $= 1800 - 1600 =$ Rs. 200

   Profit % $= \frac{200}{1600} \times 100 = 12\frac{1}{2}\%$

7. cost of 1 banana = Rs. 3.75 $\Rightarrow$ 1 dozzen = 12 $\Rightarrow$ C.P $= \frac{1600}{12} \times 3.75 =$ Rs. 500

   S.P $= \frac{900}{2} \times 1 + \frac{700}{5} \times 2 = 450 + 280 = 730 \Rightarrow\therefore$ S.P > C.P

   Profit $= 730 - 500 = 230 \Rightarrow$ Profit % $= \frac{230}{500} \times 100 = 46\%$

8. Selling Price Rs = 230 $\Rightarrow$ Profit percent = 15%

   CP $= \left(\frac{100}{100+\text{ profit}}\right) \times SP = \frac{100}{115} \times 230 = 200Rs = C.P \Rightarrow$ Profit = SP $-$ CP $= 230 - 200 =$

   30Rs

Hence the CP of the book is Rs 200& the profit on selling in it for Rs 230 it is of 30 Rupees.

9. C.P = 200Rs ⇒ Profit = 10%

Profit = 10% of C.P = $200 \times \frac{10}{100} = 20$ Rs ⇒ S.P = C.P + Profit = 200 + 20 = Rs 220

10. On selling the article for 144Rs the man loses 10%

Assume that C.P of the item = 100Rs ⇒ he loses 10% ∴ SP = 90Rs But selling given is 144Rs

∴ C.P = 100 S.P = 90Rs

? 144

on cross multiplicaton we get.

$\frac{100 \times 144}{90} = 16 \times 10 = 160$Rs

∴ C. $P = 160$ Rs But we want gain of 10%

C. P S.P. on cross multiplication

100 110

160 ? ⇒ $160 \times \frac{110}{100} = 176$Rs

∴ selling an article = Rs 176

11. Loss % = 9% then $sp_1 = 91\%$ of C.P ⇒ Profit % = 5% then $sp_2 = 105\%$ of C.P

Here 14% of C.P = 420 ⇒ 91% of C.P =? ⇒ $91 \times \frac{420}{4} = 91 \times 30 = 2730$ Rs

12. Here 1 st one profit = 2 nd one loss. ⇒ profit = loss. ⇒ SP − CP = CP − SPL

Substitute values in equation (1) ⇒ 18,000 − CP = 16,800 − CP

2CP = 34800 ⇒ CP == $\frac{34800}{2} = 17{,}400$

13. S.P. of 45 lemons is Rs 40 ⇒ loss = 20%

so C.P. of 45 lemon would be ⇒ $400 \times \frac{100}{80} = 50$

To gain 20% he should sell 45 lemons for ⇒ $50 \times \frac{20}{100} = 50 \Rightarrow 50 + 10 = 60$ Rs

Now he sells for 60Rs = 45le, oms ⇒ So for Rs 24 would sell = $\frac{24}{60} \times 45 = 2 \times 9 = 18$

18 lemons should he sell to gain 20% profit.

14. cost price = Rs $\left(\frac{100}{125} \times 8750\right) = 100 \times 70 = 7000$Rs

Let the labelled price be x repees.

Then $\frac{70}{100} \times x = 7000 \Rightarrow x = \frac{7000 \times 100}{70} = 100 \times 100 = 10{,}000$Rs

15. C.P of 100 eggs = 50 Rs

    Remaining eggs = total eggs - Broken eggs

    $= 100 - 4 = 96$ eggs

    1 dozen = 12 units $\Rightarrow \frac{96}{12}$ eggs = 8 dozens

    S.P of 1 dozen eggs = 8.50Rs

    8 dozens $= 8.50 \times 8 = 68$

    S.P = Rs 68 C.P = 50 $SP > Cp$ $\therefore$ profit will come.

    profit = S.P. - C.P = 68 − 50 = 18 Rs

    Profit % $= \frac{(\text{ profit} \times 100)}{CP} \Rightarrow \frac{18 \times 100}{50} = 36\%$

    so gain percent is 36%

16. $CP = 100 \Rightarrow$ Marked Price $= 130 \Rightarrow$ Discount Provides is $6\frac{1}{4}\% = \frac{25}{4}\%$

    $\frac{25}{4}\%$ of $130 \Rightarrow 8.125$Rs $\Rightarrow$ hence S.P becomes 121.875Rs

    hence profit = 21.87Rs which on $21\frac{7}{8}\%$ on 100Rs

17. Marked Price $= 240 + \frac{240 \times 20}{100} = 240 + 48 = 288$ Rs

    Total discount $= 288 - 264 = 24$Rs $\Rightarrow$ % of discount $= \frac{24}{288} \times 100 = 8\frac{4}{12} \Rightarrow 8\frac{1}{3}\%$

18. Expenditure on developing = 30% of 20,00,000 = 6,00,000Rs

    Total investment = 20,00,000 + 6,00,000 = Rs 26,00,000

    total profit = 35% of 26,00,000 $\Rightarrow$ 9,10,000

    S.P of each unit $= \frac{26,00,000+9,10,000}{60} = 58,500$Rs

19. Let the M.P be Rs x $\Rightarrow$ Price after discount $= x - \frac{10x}{100} = \frac{99}{10}$

    $\because \frac{9x}{10} \times \frac{108}{100} = 680.40 \Rightarrow x \times \frac{243}{250} = 680.40 \Rightarrow Rs700$

20. Selling Price $= \left(\frac{100+ \text{ profit } \%}{100}\right) \times c \cdot p$

    In case of loss $S.P = \left(\frac{100-\text{loss}\,\%}{100}\right) \times c \cdot p$

21. In case of profit $\Rightarrow C.P = \frac{S.P \times 100}{(100+ \text{ profit } \%)}$

    In case of loss $C.P = \frac{S.P \times 100}{(100-\text{loss}\,\%)}$

22. Let the CP of each mango be Re 1 , then C.P of 16 mangoes = Rs 16.

    S.P of 16 mangoes = Rs 18 .

    Gain=S.P-CP=Rs $(18 - 16) = 2$Rs $\Rightarrow$ Now Gain % $= \left(\frac{\text{Gain}}{C.P} \times 100\right)\%$

    Gain % $= \frac{2}{16} \times 100 \Rightarrow$ Gain % = 12.5%

23. C.P of the toy = Rs 250 ⇒ S.P of the toy = Rs 285

Since SP > CP : so, there is gain ⇒ Gain = S.P-CP = 285 − 250 = 35Rs

Now Gain % = $\left(\frac{\text{Gain}}{C.P} \times 100\right)\%$ ⇒ $\left(\frac{35}{250} \times 100\right) = 7 \times 2 = 14\%$ gain

24. Let the cost price of 1 st fan is Rs x. Then cost price of 2 nd fan = Rs$(2160 - x)$

It is given that

In the whole transaction the man neither gain nor loss

∴ Gain on the sale of 1 st fan=loss of sale of the 2 nd fan

⇒ 15% ofRsx = 9% ofRs $(2160 - x)$ ⇒ $\frac{15}{100} \times x = \frac{9}{100} \times (2160 - x)$

⇒ $15x = 9(1260 - x)$ ⇒ $5x = 3(2160 - x)$ ⇒ $5x = 6480 - 3x$

$8x = 6480$ ⇒ $x = \frac{6480}{8} = 810$ ⇒ C.P of the 1 st fan = Rs 810

C.P of the 2 nd fan = Rs$(2160 - x) = 2160 - 810 = $ Rs 1350

25. CP = 2500, gain % = 15% ⇒ $S \cdot P = \frac{100+15}{100} \times 2500 = \frac{115 \times 2500}{100}$ ⇒ SP = 2875

26. SP = 8400, loss = 25% ⇒ $C.P = \frac{8400 \times 100}{100-25}$ ⇒ $C.P = \frac{8400 \times 100}{75}$ ⇒ CP = 11200

27. S.P=16200, gain = 20% ⇒ $C.P = \frac{16200 \times 100}{100+20}$ ⇒ $C \cdot P = \frac{8400 \times 100}{120}$ ⇒ CP = 13500

28. Let the C.P of the article be Rs x, then gain = $\frac{1}{10}$ of $x = \frac{x}{10}$

gain % = $\left(\frac{\text{gain}}{C.P} \times 100\right)\% = \frac{\frac{x}{10}}{x} \times 100\%$ ⇒ 10%

29. C.P of 200 kg sugar = Rs(200×15)

Profit % = 5% ⇒∴ Profit = $\frac{5}{100} \times 3000 = 150$ Rs

Now SP = C · P + Profit = Rs$(3000 + 150)$Rs 3150

Hence S.P per kg = $\frac{3150}{200}$ = Rs 15.75

30. Let the C.P of the toy = Rs x. ⇒ gain % = 12% ⇒ gain = 12% of x = Rs $\frac{12x}{100} = \frac{3x}{25}$

∴ SP = CP + Gain = Rs $\left(x + \frac{3x}{25}\right) = \frac{28x}{25}$ ⇒ newgain % = 14% of x ⇒ $\frac{14x}{100}$ ⇒ $Rs\frac{7x}{50}$

∴ new S.P = Rs $\left(x + \frac{7x}{50}\right) Rs\frac{57x}{50}$

It is given that difference between new S.P. and the original S.P. is Rs.33.

∴ $\frac{57x}{50} - \frac{28x}{25} = 33 \Rightarrow \frac{57x - 56x}{50} = 33 \Rightarrow \frac{x}{50} = 33 \Rightarrow x = 33 \times 50$ ⇒ x = 1650Rs

So C.P of the toy is 1650Rs

31. Let the CP of 1 orange be Rs x. ⇒ Thus, C.P of 18 oranges = CP = 18x.

S.P of 18 oranges SP = C. P of 16 oranges ⇒ SP = 16x.

As CP is more than S.P he has suffered a loss. ⇒ loss = CP − SP = 18x − 16x

loss = 2x ⇒ loss % = $\frac{\text{loss}}{C.P} \times 100 \Rightarrow \frac{2x}{18x} \times 100\% \Rightarrow \frac{1}{9} \times 100 = 1\text{oss}\% = 11\frac{1}{9}\%$

32. CP = Rs. 250 ⇒ SP = Rs, 285 ⇒ here SP > CP ⇒ profit = S · P = C. P = 285 − 250 = 35

33. Let the C.P of 1 st TV. set be Rs x then ⇒ C.P of 2 nd TV is = Rs$(42500 - x)$

Loss of 1 st Tv set = 10% of x ⇒ $Rs\left(\frac{10}{100} \times x\right) = \frac{x}{10}$ ⇒ S.P of 2 nd Tv = C.P + Gain

$= \text{Rs}(42{,}500 - x) + \text{Rs}\frac{42500 - x}{10} = \text{Rs}(42500 - x) + \left(1 + \frac{1}{10}\right) = \text{Rs}\,42500 - x + \frac{11}{10}$

It is given that S.P of each TV set the same.

$\therefore \frac{9x}{10} = (42500 - x)\frac{11}{10} \Rightarrow 9x = (42500 - x)11 \Rightarrow 9x = 12500 \times 11 - 11x$

$9x + 11x = 42500 \times 11 \Rightarrow 20x = 42500 \times 11 \Rightarrow x = \frac{42500 \times 11}{20} = 2125 \times 11$

$= 23375Rs$

Hence the C.P of 1 st TV set = Rs 23,375

C.P of 2 nd Tv Set = Rs$(42{,}500 - 23{,}375)$ = Rs 19,125

34. S.P = Rs 67050 and loss % = 10

$\therefore CP = \frac{100}{100 - \text{loss}\,\%} \times SP \Rightarrow CP = Rs\left(\frac{100}{100 - 10} \times 67.50\right) = 75Rs$

If S.P = 82.50 then SP > CP. So gain will come

Gain S.P-C.P = 82.50 − Rs 75 = Rs 7.50 ⇒∴ gain % = $\left(\frac{\text{gain}}{CP} \times 100\right) = \frac{7.50}{7.50} \times 100 =$

10% gain % = 10%

35. a. Answer (r) ⇒ Gain % = $\frac{\text{Gain}}{CP} \times 100$ ⇒ C.P of the toy = Rs 250

S.P of the toy = Rs 285 ⇒ Since SP > CP so, there is gain given by gain = SP − CP ⇒ gain = 285 − 250 = 35Rs

b. SP = Rs100 C.P = 90Rs

SP > CP gain will come

gain = SP − CP = 100 − 90 = 10Rs ⇒ gain % = $\frac{\text{gain}}{C.P} \times 100 = \frac{10}{90} \times 100 = \frac{1}{9} \times 100 =$

$11.\frac{1}{9}\%$

c. Profit = S.P > C.P - so ⇒ Profit = Selling price - Cost price

d. loss = S. P < C.P. so ⇒ loss = cost price - selling price.

Worksheet – 4

Level – I

1) B	2) B	3) A	4) A	5) A	6) C	7) A
8) C	9) C	10) B				

1. $20 \times \frac{18}{5} = 72\text{kmph}$
2. $45 \times \frac{18}{5} = 162$ km/hr
3. $72 \times \frac{5}{18} = 20$ m/sec
4. $108 \times \frac{5}{18} = 30$ m/sec
5. $36 \times \frac{5}{18} = 10$ m/sec
6. Distance = speed × time $\Rightarrow \frac{90}{60} \times 10 = 15$ km, (time in hours)
7. $\frac{624}{6.5} = 96\text{kmph}$
8. Average speed $= \frac{90+60}{2} = \frac{150}{2} = 75\text{kmph}$
9. speed $= \frac{240}{16} \times \frac{18}{5} = 54\text{kmph}$
10. Distance $= 90 \times \frac{5}{18} \times 5 = 125$ m

Level – II

1) A	2) B	3) C	4) D	5) B	6) B	7) C
8) D	9) C	10) C	11) D	12) A	13) A	14) C
15) D	16) A	17) D	18) D	19) C	20) D	21) B
22) C	23) B	24) 50 kmph	25)1	26) A	27)A	

28) a-q, b-p, c-r, d-s

29) D	30) B	31) 3.42m/sec	32) A	33) A	34) B

35) A

1. Time $= \frac{\text{dis tance}}{\text{speed}} \Rightarrow$ Time $= \frac{20}{4} = 5$ hours
2. Speed $= \frac{\text{dis tance}}{\text{time}} = \frac{15}{2} = 7 \cdot 5$ miles / hour
3. Speed $= \left(\frac{600}{5\times60}\right)$ m/sec $= 2$ m/sec $\Rightarrow 2 \times \frac{18}{5} = \frac{36}{5} = 7 \cdot 2$ km/hr
4. Speed $= \left(\frac{5\times5}{18}\right) = \frac{25}{18} \times 15 \times 60 = 1250$ m
5. Let the speed of two trains be 7x and 8xkm/hr

   $8x = \frac{400}{4} \Rightarrow 8x = 100\ x = \frac{100}{8} \Rightarrow x = 12 \cdot 5$ km/hr

   so speed of the first train is $12 \cdot 5 \times 7 = 87 \cdot 5$ km/hr
6. Let the speed of the stream be x km/hr

   Thus downstream $= (12 + x) \Rightarrow$ and upstream $= (12 - x)$

   Suppose the distance travelled be y given $\frac{y}{12+x} = 6$ and $\frac{y}{12-x} = 9$ $[x = 2 \cdot 4$ km/hr$]$
7. Ratio of speed $= 3:4$

Distance remaining constant, the ratio of time taken = 4: 3

A taken 0 · 5 hour move then B

Hence time taken by $A = 4 \times 0 \cdot 5 = 2$ hour

8. Distance covered by Jay in 1/2hr = 4 km

Therefore, side of the square $= \frac{4}{2} = 2$ hour

Hence area = 2 × 2 = 4 square km

9. In every two minutes he is able to ascend 1 m in this fashion he ascends up to 12 m because when he reaches at the top he does not slip down. Thus up to 12 m he takes 12 × 2 = 24 min and for the last 2 m he takes 1 m

Therefore, total fine takes by him is 24 + 1 = 25min to reach the top

10. Let b and s be the speed of boat and stream respectively

As per the given condition $\frac{30}{b+s} = 2 \rightarrow (1)$

$\frac{30}{b-s} = 6 \Rightarrow$ From eq (i) and (ii) $\Rightarrow$ S = 5 and b = 10 km/hr

11. Let the length of the train be ' x ' $\Rightarrow$ So, the length of the platform = 3xm

Time taken in crossing the platform $= \frac{4x}{20}$ sec

Time taken in crossing the pole $= \frac{x}{20}$ sec $\Rightarrow \frac{x}{20} + 24 = \frac{4x}{20} \Rightarrow$ x = 160 m

12. The distance covered by the wheel in 560 revolutions = 1100 m

Hence the distance covered per revolution $= \frac{110}{560} = \frac{55}{28}$ metres

The distance covered in one revolution = circumference $= 2\pi r = 2 \times \frac{22}{7} \times r \Rightarrow$

r = 31 · 25 m

13. Let the distance to be covered by the boat when it is travelling against stream be ' x '

The boat goes down stream the river at a speed of 20 + 4 = 24

24 km/hr and up to the river at a speed 20 − 4 = 16 km/h

Since the time taken is same which is $\frac{120}{24} = \frac{x}{16}$

Therefore x = 80 km

14. Let xkm be the distance between his house and office while travelling at 24kmph, he would take $\frac{x}{24}$ hours while travelling at 25% faster speed

$$i \cdot e\ 24 + 25\%\ of\ 24 = 24 \times \left(\frac{1}{4}\right) = 30\text{kmph, he would take } \frac{x}{30} \text{ hours}$$

Now as per the problem time difference = 5 mins late 4 min early = 9

$\Rightarrow \frac{x}{24} - \frac{x}{30} = 9\text{min} = \text{X} = 18$ km

15. Ratio of speeds $= \left(\frac{550}{60} \times \frac{18}{5}\right) : \left(\frac{33}{45} \times 60\right) = 33:44 = 3:4$

16. Speed $= 5$ km/hr $= 5 \times \frac{5}{18} = \frac{25}{18}$ m/sec $\Rightarrow$ Time $= 15$ minutes

    length of the bridge = distance travelled by the man in 15 minutes $= \frac{25}{18} \times 15$

    $= 1250$ metres

17. speed from A to $B = 40$ km/hr

    speed from B to $A = 40 + 20 = 60$ km/hr

    average speed $= \frac{2\times40\times60}{40+60} = [48$ km/hr$]$

18. Total Initial bogies is $12 + 12 = 24 \Rightarrow$ Additional bogies $= 16 - 12 = 4$

24 bogies take 2 minutes $\Rightarrow$ 4 bogies will take $= \frac{2\times60}{24} \times 4 = 20$sec

19. Let the time in which he travelled on foot = x hour

    Time for travelling a bicycle $= (9 - \text{x})$hr $\Rightarrow$ Distacne = speed x time

    Total distance $= 61$ km $\Rightarrow$ So, $4\text{x} + 9(9 - \text{x}) = 61 \Rightarrow 5\text{x} = 20 \Rightarrow \text{x} = 4$

    So, distance travelled on foot $= 4(4) = 16$kms

20. Distance = Constant $\Rightarrow$ So, Speed $\propto \frac{1}{\text{time}} \Rightarrow$ Ratio of time $= 5:\frac{5}{3} = 3:1$

    $\therefore$ Ratio of speed $= 1:3 \Rightarrow 1$ unit $= 240$ km/hr $\Rightarrow 3$ units $= 240 \times 3 = 720$ km/hr

21. Average speed $= \left(\frac{2xy}{x+y}\text{ km/hr}\right) = \left(\frac{2\times25\times4}{25+4}\right)\text{km/hr} = \frac{200}{29}$ km/hr

    Distance travelled in 5 hours 48 minutes $i \cdot e = 5 + \frac{4}{5}\text{hrs} = \left(\frac{200}{29} \times \frac{29}{5}\right)\text{km} = 40$ km

    $\therefore$ Distance of the post office from the village $= \left(\frac{40}{2}\right) = [20$ km$]$

22. Speed $= \frac{200}{20}$ m/sec $= 10$ m/sec $= 10 \times \frac{18}{5} = [36$ km/hr$]$

23. Speed $= 144$kmph$144 \times \frac{5}{18} = 40$ m | sec

    Distance covered in 20 seconds $= 40 \times 20 = 800$ metres

24. Speed from A to B $= \left(250 \times \frac{2}{11}\right)\text{mph} = \left(\frac{500}{11}\right)\text{mph}$

    Speed from B to A $= \left(250 \times \frac{2}{9}\right) mph = \left(\frac{500}{9}\right) mph$

    Average Speed $= \left(\frac{2\times\frac{500}{11}\times\frac{500}{9}}{\frac{500}{11}+\frac{500}{9}}\right)\text{mph} = \left(\frac{500000}{4500+5500}\right)\text{mph} = [50\text{mph}]$

25. $\left(144 \times \frac{5}{18}\right)\text{km/hr} = 40\text{mps}$

26. $\left(20 \times \frac{18}{5}\right) m/hr = 72\text{kmph}$

27. $\left(108 \times \frac{5}{18}\right)$ km/hr = 30mps

28. a) $\left(18 \times \frac{5}{18}\right)$ km/hr = 5 m/hr

    b) $\left(10 \times \frac{18}{5}\right) m/hr$ = 36 km/hr

    c) $\left(90 \times \frac{5}{18}\right)$ km/hr = 25 m/sec

    d) $\left(50 \times \frac{18}{5}\right)$ m/hr = 180 km/hr

29. Distance = (200 × 5)km = 1000 km ⇒∴ Required speed = $\left(1000 \times \frac{3}{4}\right)$ km | hr = [750 km | hr]

30. Distance = $\left(1200 \times \frac{11}{12}\right)$ = 1100ft

31. Time = 140sec ⇒ Distance = 480 m ⇒ Speed = $\frac{\text{Distance}}{\text{Time}} = \frac{480}{140}$ = 3.42 m/s

32. Rate upstream = $\left(\frac{7}{42} \times 60\right) kmph = 10 kmph$ ⇒ Speed of stream = 3kmph

    Let speed in still water be xkm | hr then, speed upstream = (x − 3)km | hr ∴ $x - 3 = 10$(or)$[x = 13$ km | hr] ⇒ x = 13

33. Speed 126kmph ⇒ $126 \times \frac{5}{18}$ = 35 m | s 34. Speed 30 m | s ⇒ $30 \times \frac{18}{5}$ [108 km | hr]

34. Speed 10 m | s ⇒ $10 \times \frac{18}{5}$ [36 km | hr]

Worksheet – 5

Level – I

1) B	2) A	3) C	4) D	5) C	6) D	7) C
8) B	9) B	10) A				

1. Option (2) 'A' complete the work = 10 day. ⇒ in one day = $\frac{1}{10}$

   'B' complete the work = 15 days ⇒ in one day work = $\frac{1}{15}$

   borh A & B can do the work ⇒ $A + B = \frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6}$

   both A & B can do work in 6 days.

2. option (1) A Finish work in 18 days ⇒ one day work = $\frac{1}{18}$

   B can do same work in half the time taken by $A = \frac{1}{9}$ = 9 DAYS

   one day work = $\frac{1}{9}$ ⇒ both working together they can finish wok is

   $A + B = \frac{1}{18} + \frac{1}{9} = \frac{1+2}{18} = \frac{3}{18} = \frac{1}{6}$

3. Time taken by amit to do the work = 4 days

Time taken by sumit to do the work = 6 days ⇒∴ work done by Amit in 1 day $= \frac{1}{4}$

work done by Sumit in 1 day $= \frac{1}{6}$

So, work done by Amit & Sumit in 1 day $= \frac{1}{4} + \frac{1}{6} = \frac{3+2}{12} = \frac{5}{12} = 2\frac{2}{5}$ days

Hence Amit & Sumit can do the piece of work in $\frac{12}{5}$ days

4. Time taken by chotu & Nitu to plough the field together = 4 days

Field together = 4 days ⇒ Chotu & Nitu's 1 day work $= \frac{1}{4}$

Time taken by Nitu to plough the field = 6 days. ⇒ Nitu's one day work $= \frac{1}{6}$

Now, Chotu's one day work = (Chotu & Niyu's 1 day work (Nitu's 1 day's work)

$$= \frac{1}{4} - \frac{1}{6} = \frac{3-2}{12} = \frac{1}{12}$$

Hence Chotu can plough the field in 12 days.

5. 5 Men can complete a job in 8 days.

∴ 1 man can complete the job same job in $(8 \times 5) = 40$ days

So, 12 men can complete the same job in

$\frac{40}{12} = \frac{10}{3} = 3\frac{1}{3}$ days

6. Number of baskets 25

Number of days 35

As the weaving of more baskets will required more days So, it is a case of direct variation ⇒∴ Ratio of baskets numbers : Ratio of no. of days

25:10=35:x. ⇒ $\frac{25}{110} = \frac{35}{x} \Rightarrow 25x = 35 \times 110 \Rightarrow x = \frac{35 \times 110}{25} = \frac{7 \times 110}{5} = 154 \Rightarrow 154$ days

7. ∴ In 1 day 15 boys earn $= \frac{900}{5} = 180$Rs ⇒ one boy earns in one day = Rs$\frac{180}{15}$ = 12Rs

⇒ one boy earns in 7 days = Rs$(12 \times 7) = 84$Rs

then 20 boys earns in 7 days $= 84 \times 20 = 1680$ Rs

8. 20 men complete the work in 25 days

∴ 1 men can complete the work in $= 25 \times 20$ days

∴ 5 men can complete the work in $= \frac{25 \times 20}{5} = 25 \times 4 = 100$ days

9. Work of Uma & Radha for 1 day $= \frac{1}{12}$ and work of Radha for 1 day $= \frac{1}{12} - \frac{1}{18} = \frac{3-2}{36} = \frac{1}{36}$

Hence Radha will alone complete the work in 36 days.

10. Mohit & Suresh can do a piece of work in 12 days suresh can do a piece f work in30 days.

Let, mohit alone finish the work in ' x ' days.

$\frac{1}{x}+\frac{1}{30}=\frac{1}{12}\Rightarrow\frac{1}{x}=\frac{1}{12}-\frac{1}{30}\Rightarrow\frac{1}{x}=\frac{5-2}{60}=\frac{3}{60}=\frac{1}{20}=20$ days

Mohit takes 20 days to finish the work alone.

Level – II

1) 2	2) 1	3) 1	4) 3	5) 2	6) 2	7) 4
8) 4	9) 2	10) 3	11) 2	12) 3	13) 3	14) 2
15) 1	16) 3	17) 3	18) 2	19) 1		

1. Work of Ajay & Bimal for 1 day $=\frac{1}{16}$

∴ Bimal alone can complete the work in 48 days

2. Mahavir can complete in 1 day work $=\frac{1}{5}$

Ramesh can complete in 1 day work $=\frac{1}{3}$

Both can do complete the work $=\frac{1}{5}+\frac{1}{3}=\frac{5+3}{15}=\frac{8}{15}\Rightarrow\frac{15}{8}=1\frac{7}{8}$

3. Ravi & Suresh can dig work in 1 day $=\frac{1}{8}\Rightarrow$ Ravi alone can dig work in 1 day $=\frac{1}{20}$ so suresh alone can dig work in 1 day

$=\frac{1}{8}-\frac{1}{20}=\frac{5-2}{40}=\frac{3}{40}\Rightarrow\frac{40}{3}=13\frac{1}{3}\Rightarrow13\frac{1}{3}$ days can suresh alone finish the dig work.

4. Chiranji's 1 day work $=\frac{1}{15}$; Natthi in 1 day work $=\frac{1}{25}$

Sumera in 1 day work $=\frac{1}{30}$

3 persons can do that works in $=\frac{1}{15}+\frac{1}{25}+\frac{1}{30}=\frac{10+6+5}{150}=\frac{21}{150}\Rightarrow\frac{150}{21}7\frac{3}{21}=7\frac{1}{7}$ days

5. A&B can do piece of in 18 days ; A&B's one day work $=\frac{1}{18}$

B&C can do piece of work in 24 days ; B&C's one day work $=\frac{1}{24}$

C&A can do piece of work in 36 days.; C&A's one day work $=\frac{1}{36}$

Both A,B & C working together. $\frac{1}{A\&B}+\frac{1}{B\&C}+\frac{1}{C\&A}\Rightarrow\frac{1}{18}+\frac{1}{24}+\frac{1}{36}\Rightarrow\frac{4+3+2}{72}=\frac{9}{72}$

$\Rightarrow\frac{72}{9}=8$ day. $\Rightarrow 2(A+B+C)$ one day work $=2\times8=16$ days.

6. According to the question. ⇒ Let D, is the no. of days

$639\times12\times5=30\times6\times$ Days $\Rightarrow D=\frac{639\times12\times5}{30\times6}=\frac{213\times2}{6\times3}=213$ days

7. $\frac{2 \text{ persons} \times 2 \text{ days}}{2 \text{ jobs}} = \frac{100 \text{ persons} \times \text{Ddays}}{100 \text{ jobs}} \Rightarrow$ here D = no of jobs required

$(2 \times 2)100 = 100 \times 2 \times D \Rightarrow 4 \times 100 = 200D \Rightarrow 400 = 200D \Rightarrow D = \frac{400}{200} = 2$

8. $(A+B)'s 20$ days work $= \frac{1}{30} \times 20 = \frac{20}{30} = \frac{2}{3}$

Remaining work $= \left(1 - \frac{2}{3}\right) = \frac{1}{3} \Rightarrow$ Now, $\frac{1}{3}$ work is done by A in 20 days.

Whole work will be done by A in $(20 \times 3) = 60$ days.

9. No. of pages copied by (subashs prakash) ⇒ in 40 hour $= \frac{300}{40} = 7.5$ pages.

Subash copied pages in 1 hour $= \frac{50}{10} = 5$ pages

Hence, Prakash copied pages in 1 hour $= 7.5 - 5 = 2.5$

This prakash can copied 30 pages in $\frac{30}{2.5} = 12$ hours

10. Let work done by 1 man in 1 hour be m units.

Also let the work done by 1 boy in 1 hour be 6 .units

∴ work done by 5 men &2 boys in 1hr = 5 m + 2 b.

Now, as per given information,

$5M + 2B = 4 \times (M + B)$

$5M + 2B = 4M + 4B \Rightarrow M = 2B \Rightarrow M:B \Rightarrow 2:1$

∴ ratio of efficiency $= M:B = 2:1$

11. If L1. no of lamps lighted for 64. hours per day for D_1 days for Rupes R_1 and L_2 no.of lamps lighted for H_2 hrs per day D_2 days for Rupees R_2 their we have general formula this case is $\frac{L_1 H_1 D_1}{R_1} = \frac{L_2 H_2 D_2}{R_2}$

substitute the values as $H_1 = 5hrs\ H_2 = 4hrs$

$L_1 = 80$ lamps $L_2 =$ Llamps $\Rightarrow D_1 = 10$ days $D_2 = 30$ days

$R_1 = 21.25\ R_2 = 76.50 \Rightarrow \frac{80 \times 5 \times 10}{21.25} \times \frac{L \times 4 \times 30}{76.50} \Rightarrow L = \frac{80 \times 5 \times 10 \times 76.50}{4 \times 30 \times 21.25} = 20 \times 6 = 120$ lamps

L = 120 lamps

12. Let m&n be the money earned by a boy & a man in 1 day. Then, we have

$3m + 4n = \frac{2100}{7} \Rightarrow 3\text{ m} + 4\text{n} = 300 - -(1)$

$11m + 13n = \frac{8300}{8} \Rightarrow 11m + 13n = 1037.5$

Solving (1) & (2) we get $m = 50 n = 37.5$

now we need to find k where k(7 m + 9n) = 11,000

$(7 \times 50 + 9 \times 37.5) = 11{,}000 \Rightarrow 687.5 \times K = 11{,}000 \Rightarrow K = \frac{11{,}000}{687.5} = 16$

13. 1 women 1 day work $= \frac{1}{70} \Rightarrow$ 1 children 1 day work $= \frac{1}{140}$

    (5 women+10children)'s days work $= \left[\frac{5}{70} + \frac{10}{140}\right] = \frac{10+10}{140} = \frac{20}{140} = \frac{1}{7}$

    $\therefore$ 5 women &10 children will complete the work in 7 days.

14. A, B, &C together do = 1 work $\Rightarrow$ But B&C together do $= \frac{8}{23}$

    A's work $= 1 - \frac{8}{23} = \frac{15}{23} \Rightarrow \therefore$ A's shares $= \frac{15}{23} \times 520 = 15 \times 23 = 345Rs$

15. Ratio rates of working of $A\&B = 2:1 \Rightarrow$ so, ratio of time taken = 1: 2.

    $\therefore$ A's 1 day work $= \frac{1}{6}$ 2 times of $B) \Rightarrow (A + B)$'s 1 work $= \left(\frac{1}{6} + \frac{1}{12}\right) = \frac{3}{12} = \frac{1}{4}$

    $\therefore$ So A&B together can finish the work in 4 days.

16. Suppose the man works overtime for ' x ' hours.

    Now, working hours in 5 week $= 5 \times 6 \times 8 = 240$hrs

    $240 \times 40 + x \times 45 = 10{,}500 \Rightarrow 9600 + 45x = 10{,}500$

    $45\text{x} = 10{,}500 - 6900 = 900 \Rightarrow 45\text{x} = 900 \Rightarrow x = \frac{900}{45} = 20$hrs

    Total hours $= 240 + 20 = 260$hrs

17. Let the vinay take the time = A days = 30 days

    Let the vicky take the time = B days = 15 days

    Vinay+vicky together can complete 100% of one work in

    $$\frac{1}{A+B} = \frac{1}{A} + \frac{1}{B} \Rightarrow \frac{1}{A+B} = \frac{1}{30} + \frac{1}{15} \Rightarrow \frac{15+30}{30 \times 15} \Rightarrow \frac{1}{A+B} = \frac{45}{450} = \frac{1}{10}$$

    $\therefore$ A + B taking reciprocal on both sides

    $\therefore$ 100% of the work completed in 10 days.

18. Let be time taken by somu $A = 20$ days ; Let be time taken by Ramu $B = 30$ days Then no of days to complete 100% of work by Somu+Ramu together:

    $$\rightarrow \frac{1}{A+B} = \frac{1}{A} + \frac{1}{B} \Rightarrow \frac{1}{A+B} = \frac{1}{20} + \frac{1}{30}$$

    $\Rightarrow \frac{30+20}{30\times20} = \frac{50}{600} \Rightarrow$ taking both sides reciprocals $\Rightarrow A + B = \frac{600}{50} = 12$ days

    $\therefore$ 100% of the work is completed A + B by in 12 days

    50% of the work is completd in $\frac{12}{100} \times 50 = 6 \times 1 = 6$ days.

19. 1 women's 1 day work $= \frac{1}{70} \Rightarrow$ 1 Child's 1 day work $= \frac{1}{140}$.

    5 women &10 children 1 day work $\Rightarrow \left(\frac{5}{70} + \frac{10}{140}\right) = \frac{1}{7}$

    $\therefore$ 5 women &10 children will finish two work in 7 days

Level – III

20) 3 21) 4 22) 4 23) 4 24) 45min 25) 1,2 26) 1,3
27)24 28) 1 29) 2 30) 3 31) 4

20. Ram can type $\frac{32}{6}$ pages per hr. Shaam can type $\frac{40}{5} = 8$ pages per hr.

    They can type $\frac{32}{6} + 8 = \frac{40}{3}$ pages per hour. ⇒ So they will take $\left(\frac{110}{40}\right) \times 3 = \frac{33}{4}$ hout (or) 8 hours 15 minutes.

21. Given that 24 men (or) 36 women can complete a task in 29 days.

    Let M denotes men & W represent women, from the above statement we can say that. ⇒ 24 m = 36w ⇒ m = $\frac{3}{2}W$

    Now as per question how many days 8 men &17 women will take complete the same work.

    Which means we have to evaluate $8m + 17w = 8\left(\frac{3}{2}\right)w + 17w \Rightarrow 12w + 17w = 29w$

    If 36 women can complete a work 29 days, then time taken to complete the work same work will be $\frac{29\times36}{29} = 36$ days.

22. A can complete the work in 10 days ⇒ A's one day work = $\frac{1}{10}th_{\text{work}}$

    B can complete the work in 35 days ⇒ B's 1 day work = $\frac{1}{35}$ th of one work.

    C can complete one work in 70 days ⇒ C's 1 day work = $\frac{1}{70}t_{\text{of the work}}$

    Both are together, they finish the work in

    $\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) = \frac{1}{10} + \frac{1}{35} + \frac{1}{70} = \frac{7+2+1}{70} = \frac{10}{70}$ day = $\frac{70}{10}$ days = 7 days

    The work will get completed in 7 days.

23. Given that $A + B = 12$ days ⇒ A is thrice as efficient as B.

    ⇒ A takes × days to complete the work.

    While B takes $3x$ days to complete the work. ⇒ $\frac{1}{A+B} = \frac{1}{A} + \frac{1}{B} \Rightarrow \frac{1}{12} = \frac{1}{x} + \frac{1}{3x}$

    $\frac{1}{12} = \frac{3+1}{3x} \Rightarrow \frac{1}{12} = \frac{4}{3x} \Rightarrow 12 = \frac{3x}{4} \Rightarrow x = 12 \times \frac{4}{3} = 4 \times 4 = 1$ day

    Thus A takes 16 days to complete the work B take 16 × 3 = 48 days to complete work.

24. Distance to be covered = 1500 m × 4 = 6000 m = 6 km

    Speed = 8kmn ⇒ Time = $\frac{\text{dis tance}}{\text{speed}} = \frac{6}{8} = \frac{3}{4}$hrs = 45 minutes

25. Answers : (1) (2) & (3) ⇒ Total days taken $= \frac{100}{40} \times 12 = 30$ days

26. Answer : (1) & (3) A completes the work in 42 days

    A's oneday work be $\frac{1}{42}$ of the work

    B is twice as efficient as A. Which means B will take only half of the time taken by A.

    B will complete the works in 21 days. ⇒ B's one dau work $= \frac{1}{21}$ of the work.

    working together, in one day $\frac{1}{42} + \frac{1}{21} = \frac{1+2}{42} = \frac{3}{42} = \frac{1}{14}$

    of the work will get. The will complete in 14 days

27. Option (2) (3) & (4) : Let B takes ' x ' days to complete the work. 'A. takes $2x$ days to complete the work

    B's 1 day work $= \frac{1}{x} \Rightarrow$ A's 1 day work $= \frac{1}{2x}$

    $(A+B)'s1$ day work $= \frac{1}{x} + \frac{1}{2x} = \frac{1}{8}. \Rightarrow \frac{(2x+1)}{2x} = \frac{1}{8} \Rightarrow \frac{3}{2}x = \frac{1}{8}$

    $3 \times 8 = 2x \Rightarrow\Rightarrow 2x = 24 \Rightarrow x = 12$ days ⇒ A takes $2 \times$ days $= 2 \times 12 = 24$ days

28. As the rule of men, work, days & hours relation.

29. As for the formula. 30. As for the formula. 31. As for the formula.

6. EXPONENTS & POWERS

Worksheet – 1

Level – I

1) B	2) D	3) C	4) C	5) D	6) A	7) B
8) A	9) D	10) A				

Level – II

1) A	2) B	3) A	4) D	5) 16	6) A	7) B
8) B	9) C	10) B	11) 9	12) A	13) B	14) 12
15) 49	16) A	17) A	18) B	19) B	20) -1	21) A,B
22) B,D	23) A	24) C	25) A	26) D	27) a-r;b-s;c-q;d-p	
28) A,C	29) C	30) C	31) C	32) A,C	33) A,B,C	34) A,D

Worksheet – 2

Level – I

1) C	2) B	3) B	4) B	5) C	6) A	7) A
8) A	9) A					

Level – II

1) A	2) D	3) C	4) 18	5) B	6) B	7) A
8) A	9) B	10) B	11) A	12) A	13) A	14) B
15) A	16) D	17) B	18) B	19) C	20) A	21) A,B
22) C,D	23) A	24) A	25) A	26) A	27) a-q;b-p;c-s;d-t	
28) A,D	29) A,B,C,D	30) A	31) B	32) B	33) A,B	34) C,D
35) A,B	36) a-r;b-p;c-t;d-q					

SOLUTIONS

Level – II

11. $x^{mn-mp} \times x^{np-nm} \times x^{pm-pn} \Rightarrow x^{man-mpp+pp-pmn+pxn-phr} \Rightarrow x^0 = 1$

12. $a = \left(\frac{-9}{5}\right)^5 \Rightarrow a^{-1} = \left(\frac{-9}{5}\right)^5$

13. $5^{x+3} = 5^{2(3x-4)} \Rightarrow 5^{x+3} = 5^{6x-8} \Rightarrow x + 3 = 6x - 8 \Rightarrow 5x = 11 \Rightarrow x = 11/5$

15. $2^x \times \frac{1}{2} + 2^x \times 2 = 320 \Rightarrow 2^x\left(\frac{1}{2} + 2\right) = 320 \Rightarrow 2^x \times \frac{5}{2} = 320$

$$2^x = 64 \times 2 \Rightarrow 2^x = 2^7 \Rightarrow x = 7$$

16. $2^2 \times 2^{3\,m} = 2^5 \Rightarrow 2 + 3\,m = 5 \Rightarrow 3\,m = 3 \Rightarrow m = 1$

$1 + 1 + - - - - + 2014 \text{ (times)} = 2014$

17. $(5^{2x}) = 5^{3y} \Rightarrow 2x = 3y \Rightarrow x : y = 3 : 2$

18. $a^{\frac{xyz}{xy}} \cdot a^{\frac{xyz}{yz}} \cdot a^{\frac{xyz}{zx}} = a^{x+y+z}$

19. $2^x \times 2^{\frac{3}{5}} = 2^{\frac{1}{5}} \Rightarrow x + \frac{3}{5} = \frac{1}{5} \Rightarrow x = \frac{-2}{5}$

20. $x^6 = 65 - 1 = 64 \Rightarrow x^6 = 2^6 \Rightarrow x = 2 \Rightarrow x^4 = 2^4 = 16$

24. $x^{a+b-} = x^{a+b+c}$

25. $(x^3x^2)^4 = x^{12}x^8 = x^k x^1 \Rightarrow k + 1 = 12 + 8 = 20$

26. $3^{x+8} = 3^{6x+3} \Rightarrow x + 8 = 6x + 3 \Rightarrow 5x = 5 \Rightarrow x = 1$

28. $(256)^{0.16+0.09} = (256)^{0.25} = (4^4)^{\frac{1}{4}} = 4$

29. $4^{2^{2^1}} = 4^4$

30. $2^a = 64; 4^b = 64 \Rightarrow 2^a = 2^6; 4^b = 4^3 \Rightarrow a = 6; b = 3 \Rightarrow \frac{2}{6} + \frac{1}{3} = \frac{2}{3}$

31. $5^{\frac{1}{4}} \times (5^3)^{\frac{1}{4}} = 5^{\frac{1}{4}+\frac{3}{4}} = 5$

32. $(4^3)^{\frac{2}{6}} = 4$

33. $x^{a^2-b^2} \times x^{b^2-c^2} \times x^{c^2-a^2} \Rightarrow x^0 = 1$

34. $(49)^{\frac{1}{2}} = 7$

35. $2^{10+n} \times 2^{6n-10} \times 2^{-4n-1} \times 2^{1-3n}$

$2^{10+n+6n-10-4n-1+1-3n}$

$2^{7n-7n-1+1} = 2^0 = 1$

Worksheet – 3

Level – I

1) B	2) B	3) A	4) B	5) A	6) A	7) B
8) C	9) C	10) A				

Level – II

1) B	2) A	3) B	4) D	5) B	6) 162	7) D
8) B	9) C	10) C	11) A	12) C	13) A	14) B
15) A	16) D	17) A	18) D	19) A	20) A,C	21) A,B,C
22) A	23) A	24) B	25) A	26) a-q;b-p;c-s;d-r,t		27) A,B
28) A,C	29) D	30) B	31) A	32) B	33) C,D	34) A,D

35) a-q;b-p;c-s;d-r

SOLUTIONS:

Level – II

11. $(3^3)^{\frac{-2}{6}} \times (4^3)^{\frac{2}{3}} \times (3^4)^{\frac{-3}{12}} = \frac{1}{3} \times 16 \times \frac{1}{3} = \left(\frac{4}{3}\right)^2$

12. $\frac{(-30)^{-1}}{x} = (6)^{-1} \Rightarrow x = \frac{(-30)^{-1}}{(6)^{-1}} \Rightarrow \frac{-6}{30} = \frac{-1}{5}$

13.

14. $2^{10+n+6n-10-4n-1-3n+1} = 2^0 = 1$

15. .

16.

17. $\frac{(243)^{0.2}}{(7)^{0.25} \times (7)^{0.15} \times (7)^{0.6}} = \frac{(3^5)^{0.2}}{7^{0.25+0.15+0.6}} = \frac{3^1}{7^1} = \frac{3}{7}$

18. $2^n \times 2^4 - 2^n \times 2^2 = 3 \Rightarrow 2^n(16-4) = 3 \Rightarrow 2^n = \frac{3}{12} \Rightarrow n = -2$

19. $(6^3)^{2/3} + (4^4)^{3/4} + (2^5)^{1/5} = 36 + 64 + 2 = 102$

20. $2^{7+2a} \div 3^{3a+11} \Rightarrow \frac{2^{7-8}}{3^{-12+11}} = \left(\frac{2}{3}\right)^{-1} = \frac{3}{2}$

21. $5^x = 5^3\ 6^y = 6^2\ 7^z = 7^1$

$x = 3\ y = 2\ z = 1$

$(3 + 2 + 1)^2 = 6^2 = 36$

32. $x^2y + y^2x + y^2z + yzz^2 + zx^2 + z^2x$

$9 \times 2 + 4 \times 3 + 4 \times 1 + 2 \times 1 + 1 \times 9 + 1 \times 3 \Rightarrow 18 + 12 + 4 + 2 + 9 + 3 = 48 = 3y^{y+z}$

33. $3^{2n} \times 3^2 \times \frac{1}{3^m} \times \frac{1}{3^{+3n}} \times \frac{3^{2\ m}}{3^3} = \frac{1}{27}$

$$\frac{3^{2n} \times 3^{2m}}{3^m \times 3^{3n}} = \frac{3}{27} \Rightarrow \frac{3^m}{3^n} = \frac{1}{9} \Rightarrow 3^{m-n} = 3^{-2} = m - n = -2$$

34. m-terms

$$\frac{20^m \times 12^m \times 15^m}{16^m \times 25^m \times 9^m} = \left(\frac{20 \times 12 \times 15}{16 \times 25 \times 9}\right)^m = 1$$

n-terms $\Rightarrow \frac{4^n \times 12^{-n} \times 15^n}{5^n} = \left(\frac{4 \times 15}{5 \times 12}\right)^n = 1$

Constant: $\frac{20^{-1} \times 15^{-2}}{9^{-1}} = \frac{9}{20 \times 15 \times 15} = \frac{1}{500}$